REGRET IN RUNTIME ANALYSIS OF BLACK-BOX ALGORITHMS

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Abstract

TODO: Add abstract after completion of report

Declaration

No portion of the work referred to in this report has been submitted in support of an application for another degree or qualification of this or any other university or other institute of learning.

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Abbreviations and Acronyms

B | C | D | E | G | H | J | M | N | P | R | S | T | Y

B

BBC Black-Box Complexity. 20

BBO Black-Box Optimization. 12, 13, 16, 18, 20, 21, 25, 30, 41

BO Bayesian Optimization. 32

C

CI Continuous Integration. 14, 46

CLI Command-Line Interface. 42, 46

CNF Conjunctive Normal Form. 24

CR Cumulative Regret. 33, 37

D

DFO Derivative-Free Optimization. 19

DPI Dots per inch. 45

E

EA Evolutionary Algorithms. 12, 18, 20, 21, 22, 23, 25, 26, 31, 36, 37, 40, 51, 52, 53, 54, 63, 66

G

GA Genetic Algorithms. 23

H

HIFF Hierarchical If-and-Only-If. 23, 24, 50, 51, 52

J

JSON *acr. JavaScript Object Notation.* An open standard file format and data interchange format that uses human-readable text to store and transmit data objects consisting of name–value pairs and arrays (or other serializable values).. 3, 15, 42, 43, 44

M

MAB Multi-Armed Bandits. 32

MaxSAT Maximum Satisfiability problem. 24

N

NFL No-Free-Lunch. 25, 26

P

PDF Portable Document Format. 15, 45

R

RLS Randomized Local Search. 20, 29, 30, 44, 49, 51, 52, 53, 67

RSH Randomized Search Heuristic. 12, 25, 26

S

SA Simulated Annealing. 12, 20, 26, 27, 28, 51, 66

SAT Satisfiability problem. 24

SR Simple Regret. 34

T

TDD Test-Driven Development. 46

TTFO Time to First Optimum. 13, 45, 48, 51, 53

Y

YAML *acr. YAML Ain't Markup Language.* A human-readable data serialization language, commonly used for configuration files.. 13, 42, 43, 46

Chapter 1

Introduction

1.1 Motivation

Across engineering, the natural sciences, and machine learning, a recurring class of optimization problems resists conventional analytical treatment. The objective function may arise from a physical experiment, a high-fidelity simulation, or a complex computational pipeline whose internal structure is either inaccessible or too intricate to exploit. In such settings, the practitioner's only recourse is to query the objective at a candidate solution and observe the resulting value; no gradient, curvature, or closed-form expression is available [Liu+20]. Problems of this kind are collectively termed *Black-Box Optimization (BBO)* problems. Industrial process optimization, drug-design pipelines, and the tuning of machine-learning models [BAH21; Bis+17] belong to this class of problems.

Without structural access, practitioners turn to general-purpose *Randomized Search Heuristic (RSH)*s: stochastic algorithms such as Evolutionary Algorithms (EA) and Simulated Annealing (SA) that search by iterative perturbation and selection [AAS18; RDD23]. The standard theoretical question for such algorithms is the number of evaluations needed to first sample the optimum. This single metric (the expected runtime) has dominated the field for decades and underpins a rich body of complexity results [DJW02; DN20; Len19].

However, expected runtime is a blunt instrument. It says nothing about solution quality *before* the optimum is reached. In practice, budgets are finite, runs are interrupted, and the best-so-far solution is what actually gets used [JZ14]. An algorithm that stagnates and then gets lucky looks identical, by this measure, to one that makes steady progress throughout. This gap has been noted explicitly in the literature: Pinto and Doerr [PD18] call for *combined anytime measures* from machine learning, naming regret as a natural candidate while acknowledging that a satisfying formulation remains open.

This project answers that call. By connecting the *regret* framework from online learning [CL06; LS20] to the *runtime profile* framework from evolutionary computation [PD18], a unified anytime measure is established — one that captures solution quality across the *entire* run, not only at its end. A bespoke experimental framework is developed to validate this connection and to investigate, across a broad range of benchmark problems and algorithm families, what regret trajectories reveal about landscape structure that classical measures cannot.

1.2 Aims and Objectives

Broader Aims	Specific Objectives
Aim 1. *Consolidate the theoretical foundations of BBO, including the standard performance measures and their limitations.*	1. Formalize the expected runtime, fixed-budget, and fixed-target frameworks. 2. Identify the shared limitation: the absence of a combined anytime performance measure. 3. Review the regret vocabulary from online learning and situate it relative to BBO complexity theory.
Aim 2. *Explore a viable relationship between the canonical measure of BBO algorithms and regret.*	1. Define the distributional runtime profile $\mathcal{P}(v,t) = \mathbb{P}(\tau_v > t)$. 2. Prove the tail-sum regret representation identity (Theorem 3). 3. Derive the asymptotic corollary connecting long-run regret to the expected hitting times $\mathbb{E}[\tau_v]$.
Aim 3. *Develop a reproducible experimental framework suited to trajectory-level regret analysis.*	1. Implement the `regret` framework with declarative YAML-driven experiment dispatch. 2. Support full trajectory recording, regret metric computation, and runtime profile analysis. 3. Validate the implementation via a test-driven development discipline with enforced coverage distribution.
Aim 4. *Investigate empirically how regret trajectories reflect landscape structure across a broad benchmark suite.*	1. Verify the tail-sum identity numerically on tractable benchmarks. 2. Characterize the diagnostic signatures of barrier-crossing, epistatic ruggedness, and hierarchical difficulty in cumulative regret trajectories. 3. Identify conditions under which classical measures (TTFO, $P(\text{opt})$) are uninformative and regret trajectories remain structured.

Table 1.1: Project aims and specific objectives.

1.2.1 Success Criteria

Following this project's intended aims (Table 1.1), the *success criteria* for these achievements are quantified in Table 1.2.

Area	Criteria
THEORY	The tail-sum identity (Theorem 3) must be both proved rigorously *and* corroborated numerically to within Monte-Carlo sampling error on at least two benchmark functions. Proof alone is insufficient: if the identity cannot be verified in the finite, discrete experimental setting, it provides no practical basis for the empirical analyses that follow. Agreement between trajectory-derived and profile-derived expected cumulative regret curves procured from experiments constitutes the verification.
EMPIRICS	Cumulative regret trajectories must yield qualitatively distinct signatures across at least three landscape archetypes (for instance, unimodal, barrier, and deceptive) that are not separated by expected runtime (time to optimum) alone. This is the operational test of the project's central claim that regret provides strictly more diagnostic information than classical measures. Without this, the theoretical contribution, though formally correct, would have no demonstrated practical utility.
SOFTWARE	The `regret` framework must execute all 14 experimental suites reproducibly and maintain a minimum of 50% branch coverage enforced in CI. This threshold is the evidentiary floor for the empirical contribution: results obtained from an unvalidated implementation cannot support claims about algorithm behavior, and reproducibility is a precondition for any comparative conclusion.

Table 1.2: Success criteria for the project

1.3 Contributions

This project makes the following principal contributions, as summarized in Table 1.3.

Area	Contributions
THEORY	Established a formal connection between expected incumbent cumulative regret and the distributional runtime profile of Pinto and Doerr [PD18] (Theorem 3), yielding the *combined anytime measure* identified as an open problem in the evolutionary computation literature. Corollary 4 relates infinite-horizon regret to the sum of expected first hitting times over all fitness levels, bridging expected-runtime analysis and regret-based evaluation. The results hold for any randomized black-box algorithm on any finite pseudo-Boolean domain.
EMPIRICS	Introduced a structured suite of 13 experiments that numerically validates the tail-sum identity (Suite 08) and studies how regret trajectories reflect landscape structure, including barrier crossings (Suites 04–06, 09–10), epistatic ruggedness (Suites 03, 12), hierarchical difficulty (Suite 07), and scaling behavior (Suite 13). The results show that cumulative regret captures diagnostic information about trapped phases and saturation that fixed-budget and expected-runtime measures miss when $P(\text{opt}) \approx 0$.

Area	Contributions
SOFTWARE	Developed `regret`, a declarative, reproducible, and parallelized experimental framework for pseudo-Boolean and combinatorial optimization problems, with support for trajectory recording, regret metrics, runtime-profile analysis, and structured output in JSON and PDF formats.

Table 1.3: Main contributions from the project.

1.4 Report Structure

This report is structured as follows:

Background [Chapter 2] compiles the necessary knowledge sufficient to grasp the intent and work presented in the later texts.

Runtime Profiles and Anytime Performance [Chapter 3] delves into the core theory developed this project, coupling a measure of regret with the metric of 'Runtime Profiles'.

Supplementary Software Design [Chapter 4] documents the novel, supplementary experimental frameworks developed for this project. This software facilitated the experimentation and exploratory analysis of various standard BBO problems

Methodology [Chapter 5] inscribes the exploratory and verification (based on concepts in Chapter 3) experiments coordinated in this project; emphasis is provided on specific rationale and decisions taken for motivating the exploration.

Evaluation and Discussion of Results [Chapter 6] confers the results gathered from the experiments conducted. Pointwise regret metrics extracted from the experiments (against known optimums) are analyzed qualitatively and quantitatively (especially for the verification experiments mentioned earlier).

Conclusion and Future Work [Chapter 7] collates the substance, contributions, and achievements of the project. Reflections and potential future directions beyond the seminal work in this project are mapped out.

Chapter 2

Background

This chapter establishes the conceptual vocabulary required to understand, motivate, and situate the contributions of this project. The subject is the regret-based analysis of randomized search heuristics on pseudo-Boolean black-box optimization problems. Four bodies of knowledge bear on this subject: the structure of the optimization problems themselves, the algorithms used to solve them, the performance measures used to evaluate those algorithms, and the regret framework imported from online learning to address the limitations of classical measures. These are introduced in that order.

2.1 Computational Problems

2.1.1 Optimization Problems

An *optimization problem* is a formal specification of the task of finding a candidate solution that is optimal with respect to a given objective function. The optimization can be minimizing or maximizing; this project adopts the latter convention. Given a set of candidate solutions $\mathcal{X}$ and an objective function $f : \mathcal{X} \to \mathbb{R}$, the goal is to identify

$$x^* \in \operatorname*{arg\,max}_{x \in \mathcal{X}} f(x).$$

A further distinction concerns the role of randomness in the problem formulation. In a *deterministic* formulation, both the problem instance and every algorithm operating on it are fully specified without any probabilistic component. In a *stochastic* formulation, the instance itself, the objective evaluations, or the algorithm may involve randomness. *Noisy* objective functions, where repeated evaluation of the same candidate solution may return different values, are a canonical example and arise naturally in simulation-based and real-world settings [Liu+20]. All problems studied in this project are deterministic, though the algorithms applied to them are randomized; this distinction is elaborated in Section 2.2.1.

2.1.2 Search Space

2.1.2.1 Structure of the Search Space

The *search space* $\mathcal{X}$ is the set of all candidate solutions considered by an algorithm. Search spaces may be *finite* or *infinite*, and *discrete* or *continuous*.

Discrete combinatorial search spaces, in particular the Boolean hypercube $\{0,1\}^n$, are central

to the theoretical analysis of evolutionary algorithms (EA) and other randomized search heuristics [DJW02; HS05b] used in theoretical BBO analysis. Continuous spaces $\mathbb{R}^d$ arise in numerical optimization [NW06], and mixed spaces containing both discrete and continuous variables occur in practice but are considerably harder to analyze [AH17].

The structure of the search space, and in particular how neighboring solutions relate to one another, fundamentally influences the difficulty of optimization. This is encoded by a *neighborhood function* $\mathcal{N} : \mathcal{X} \to 2^{\mathcal{X}}$, assigning to each candidate solution the set of solutions reachable in a single step [HS05a]. For the Boolean hypercube $\{0,1\}^n$, the standard neighborhood consists of all strings at Hamming distance one[1].

This project is concerned exclusively with *pseudo-Boolean optimization*: problems in which the search space is the Boolean hypercube $\mathcal{X} = \{0,1\}^n$ and the objective function takes non-negative real values. A candidate solution is represented as a *bitstring*, a sequence of bits $x = (x_1, \ldots, x_n) \in \{0,1\}^n$, and the integer n is referred to as the *problem dimension*. The hypercube has cardinality 2^n, so exhaustive enumeration of all candidate solutions is intractable for any practically relevant n.[2]

2.1.2.2 Constraints and Feasibility

In a *constrained* problem, the *feasible region* $\mathcal{F} \subseteq \mathcal{X}$ is defined by conditions a valid solution must satisfy; an *unconstrained* problem places no restriction beyond membership in $\mathcal{X}$ [NW06]. At times, the constraints themselves can be black-box. All pseudo-Boolean problems studied in this project are unconstrained.

2.1.3 Objective Functions

The geometric structure of the objective function determines both which algorithms are applicable and what convergence guarantees are attainable. A central organizing principle is that access to higher-order analytical information about f, i.e., gradients, curvature, or structural properties such as convexity, generally enables stronger theoretical guarantees and more efficient algorithms; restricting this access, as in the black-box setting, correspondingly limits what can be guaranteed.

2.1.3.1 Oracle-Based Computation

An *oracle* is an abstract computational device that answers queries about a problem instance. In optimization, the most common oracle is the *function evaluation oracle*, which, given a query point $x \in \mathcal{X}$, returns the objective value $f(x)$ (or a noisy estimate thereof). The algorithm interacts with the oracle in a *query-response* model: it submits a query and receives a response before deciding on the next query.

[1] *Hamming distance* between two bitstrings of equal length is the number of positions at which the corresponding bits differ

[2]

Remark 1 (Objective function convention)**.** *Without loss of generality, f is taken to be non-negative throughout this project. Any objective function with co-domain $\mathcal{T} \subseteq \mathbb{R}$ can be shifted via the affine map*

$$g : \mathbb{R} \to \mathbb{R}_0^+, \qquad g(x) = x - \min_{y \in \mathcal{X}} f(y),$$

which is well-defined whenever f attains its minimum on $\mathcal{X}$ and preserves the ordering of all objective values by construction.

2.1.3.2 Zeroth-, First-, and Second-Order Methods

Optimization algorithms are classified by the order of analytical information they require. *Zeroth-order* methods use only evaluations of $f(x)$; *first-order* methods additionally access the gradient $\nabla f(x)$; *second-order* methods further access the Hessian $\nabla^2 f(x)$. Access to higher-order information generally enables stronger convergence guarantees and more efficient algorithms [NW06], but requires corresponding structural assumptions about f. Formal definitions and convergence rates for each tier are given in Appendix B.1.

This project operates exclusively in the zeroth-order regime. When the internal structure of f is inaccessible or analytically intractable, function evaluations are the only information available. Hence, these problems are often referred to as *Derivative-Free Optimization (DFO)* problems. These conditions lay ground for the black-box model formalized in Section 2.2.

2.2 Black-Box Optimization

The project is concerned with algorithms that interact with the objective function exclusively through a query oracle, the *black-box* setting. This section formalizes that model, establishes the complexity measures used to evaluate algorithms within it, and introduces the benchmark problems and algorithm families that are studied throughout the rest of the report.

2.2.1 Algorithms

An *algorithm* is a finite, unambiguous sequence of instructions that transforms a given input into a desired output within a bounded number of steps [HS05a]. In the context of optimization, the input is a problem instance and the output is a candidate solution together with its objective value; algorithm quality is judged both by the solutions returned and the computational resources consumed.

Adaptive Algorithms. An algorithm is *adaptive* if each action at step t is chosen as a function of the history of prior queries and their responses, $(x_1, f(x_1)), \ldots, (x_{t-1}, f(x_{t-1}))$; a *non-adaptive* algorithm such as pure random sampling fixes all query points in advance and serves as a natural complexity baseline [DJW06]. All algorithms studied in this project are fully adaptive.

Deterministic and Randomized Algorithms. A *deterministic* algorithm produces the same sequence of operations and the same output on every execution for a given input. A *randomized* algorithm, by contrast, makes probabilistic choices during execution, so its behavior on a fixed input may vary across runs. In the black-box optimization setting, randomization is not merely a convenience but a necessity: the No-Free-Lunch theorems imply that no deterministic algorithm can perform well uniformly across all problem instances (See 2.2.6.1, Wolpert and Macready [WM02]), and randomized search heuristics derive their practical value from their ability to escape local optima through stochastic perturbation [DN20; HS05a]. The theoretical analysis of such algorithms therefore concerns expected quantities (expected runtime, expected solution quality) rather than worst-case deterministic guarantees.

2.2.2 The Black-Box Model

In the *black-box optimization* model, the algorithm has no access to the analytical form of the objective function. It interacts with the objective exclusively through an oracle that, upon receiving a query $x \in \mathcal{X}$, returns $f(x)$ (or a noisy estimate thereof) [DJW06; DN20; Len19]. The total number of such interactions constitutes the *query complexity* of the algorithm, formalized in Section 2.2.3.2 [NW06; DJW06]. The black-box model is appropriate whenever the objective arises from a simulation, physical experiment, or any process whose internal structure is inaccessible or too complex to exploit analytically. In fact, it has been observed that BBO methods are sometimes applied due to their ease of use rather than necessity [BAH21].

In contrast, in *white-box optimization*, some information about the problem instance is made available to the algorithm (i.e., properties described in Subsubsection 2.1.3.2, Subsection 2.2.5).

2.2.2.1 Taxonomy

Three models are relevant to this project. In the **unrestricted model**, the algorithm has arbitrary memory and full access to function values, yielding the weakest restriction and smallest complexity [DJW06]. In the **unary unbiased model**, variation operators are invariant under permutations and bitwise complementations of the search space [LW10]; both RLS and the $(1+1)$ EA fall into this class, making its BBC the tightest applicable lower bound for these algorithms. SA fits neither model cleanly (it exploits absolute fitness differences rather than rankings) so its performance is studied empirically.

2.2.2.2 Yao's Minimax Principle

Lower bounds are proved via Yao's minimax principle [Yao77]. Let $\mathcal{I}$ be the set of problem instances, $\mathcal{A}$ the set of deterministic algorithms, and $T(I, A)$ the query cost (number of oracle calls) of algorithm $A \in \mathcal{A}$ on instance $I \in \mathcal{I}$. For any distribution p over instances and q over deterministic algorithms,

$$\min_{A \in \mathcal{A}} \mathbb{E}_{I \sim p}[T(I, A)] \leq \max_{I \in \mathcal{I}} \mathbb{E}_{A \sim q}[T(I, A)].$$

A lower bound follows by exhibiting a distribution over instances on which every deterministic algorithm is slow. This parallels the minimax regret framework (Section 2.4.2.2), replacing the hitting criterion with a trajectory-level measure. The principle extends to memory-restricted and elitist models via a suitable superset construction [DL15] (see Appendix C).

2.2.3 Complexity Measures

2.2.3.1 Classical Complexity: Time and Space

The classical measures of algorithmic efficiency are *time complexity* (the number of elementary operations performed as a function of input size) and *space complexity* (the amount of memory consumed). Both are typically expressed using asymptotic notation: $O(\cdot)$, $\Omega(\cdot)$, and $\Theta(\cdot)$ describe upper, lower, and tight bounds, respectively, on resource usage as the input size n grows.

2.2.3.2 Information Complexity: Query and Oracle Complexity

In BBO, the internal structure of f is inaccessible, so the relevant measure of complexity is not the number of arithmetic operations but the number of oracle queries required to solve a problem. The *query complexity* of an algorithm $\mathcal{A}$ on a problem class $\mathcal{F}$ is the number of calls to the function-evaluation oracle needed to identify a solution meeting the target criterion. The *black-box complexity* of a problem class is the minimum query complexity over all algorithms, formalizing the information-theoretic difficulty of the problem when only objective values are observable [DJW06]. Lower bounds on black-box complexity establish that no algorithm can solve the problem in fewer queries, regardless of its computational power; upper bounds are witnessed by constructive algorithms. This duality mirrors the classical worst-case analysis of deterministic algorithms but is applied to the query-response model introduced in Section 2.1.3.1.

2.2.3.3 Expected vs. Worst-Case Analysis

For deterministic algorithms, worst-case analysis over all possible inputs is the standard framework. For randomized algorithms, and in particular for the randomized search heuristics studied in this project, the primary measure is the *expected runtime*: the mean number of function evaluations until a target solution is first found, averaged over the internal randomness of the algorithm [DJW02; DN20]. Expected runtime analysis is complemented by tail bounds (probabilistic guarantees that the runtime does not greatly exceed its expectation) derived via tools such as Markov's inequality and concentration arguments [DN20]. Worst-case analysis over problem instances, by contrast, motivates the minimax framework and Yao's minimax principle, which lower-bounds the expected complexity of any randomized algorithm by exhibiting a hard distribution over instances [DJW06] (Subsubsection 2.2.2.2).

2.2.4 A Few Pseudo-Boolean Problems

The *pseudo-Boolean* setting considers functions $f : \{0,1\}^n \to \mathbb{R}$ defined on binary strings of length n. Several benchmark functions are widely used to study and compare the behavior of randomized search heuristics, each encoding a different type of difficulty. [Len19; DJW06]

2.2.4.1 OneMax

The OneMax function counts the number of one-bits in a bit string:

$$\textsc{OneMax}(x) = \sum_{i=1}^{n} x_i, \qquad x \in \{0,1\}^n.$$

The unique global maximum is $x^* = 1^n$ with $\textsc{OneMax}(1^n) = n$. OneMax is the canonical baseline benchmark: the landscape is unimodal and every improving move increases the function value by exactly one, providing a clear gradient for local search methods. The expected runtime of the $(1+1)$ Evolutionary Algorithms (EA) on OneMax is $\Theta(n \log n)$. [DJW02]

2.2.4.2 LeadingOnes

The LeadingOnes function counts the length of the maximal leading prefix of ones:

$$\textsc{LeadingOnes}(x) = \sum_{i=1}^{n} \prod_{j=1}^{i} x_j.$$

Equivalently, $\textsc{LeadingOnes}(x)$ equals the index of the first zero bit minus one, or n if no zero exists. The unique global optimum is again 1^n. Because progress requires the leading bit to be set before any subsequent bit can contribute, LeadingOnes is significantly harder than OneMax: the expected runtime of the $(1+1)$ EA is $\Theta(n^2)$. [DJW02]

2.2.4.3 Jump

The Jump function with parameter $k \in \{1,\ldots,n\}$ coincides with OneMax on bit strings with at most $n-k$ ones and on the global optimum 1^n, but introduces a region of reduced fitness for strings with between $n-k+1$ and $n-1$ ones. This creates a "gap" of width k that algorithms relying on single-bit improvements cannot cross: the set of strings with exactly $n-k$ ones constitutes a local optimum of standard local search, and reaching the global optimum 1^n requires flipping all k remaining zero bits simultaneously. For the $(1+1)$ EA with standard bit mutation at rate $1/n$, the probability of a specific k-bit flip is $\Theta(n^{-k})$, giving an expected runtime of $\Theta(n^k)$ on $\textsc{Jump}_k$.

Formally, $\textsc{Jump}_{k,n}\colon \{0,1\}^n \to \mathbb{R}$ is defined as

$$\textsc{Jump}_{k,n}(x) = \begin{cases} k + \sum_{i=1}^{n} x_i & \text{if } \sum_{i=1}^{n} x_i \le n-k \text{ or } \sum_{i=1}^{n} x_i = n, \\ n - \sum_{i=1}^{n} x_i & \text{otherwise.} \end{cases}$$

Droste, Jansen, and Wegener [DJW02] proved that the expected running time of the $(1+1)$ EA on $\textsc{Jump}_{k,n}$ is $\Theta(n^k + n\log n)$ for $k \in \{1,2,\ldots,n\}$, which simplifies to $\Theta(n^k)$ for $k \ge 2$.

2.2.4.4 BinaryValue [BinVal]

The BinaryValue (BinVal) function interprets a bit string as a binary integer by weighting bits exponentially by position:

$$\textsc{BinVal}(x) = \sum_{i=1}^{n} 2^{i-1} x_i.$$

The global optimum is 1^n with value $2^n - 1$. The exponential weighting means that the most significant bit contributes more than the combined value of all less significant bits, making BinVal highly structured.

2.2.4.5 Trap

The Trap function is a piecewise-linear *function of unitation* formally analyzed for deception conditions by Deb and Goldberg [DG93]. Its value depends only on the number of ones $u = \sum_{i=1}^{n} x_i$, not

on their positions, and with slope-change location $z = n - k$ it is defined as

$$\textsc{Trap}_k(x) = \begin{cases} \frac{n}{z}(z-u) & \text{if } u \le z, \\ \frac{n}{k}(u-z) & \text{otherwise.} \end{cases}$$

For $u \le z$ the gradient points toward the all-zeros attractor 0^n; only past the slope change does fitness increase toward the global optimum 1^n. Every improving move for a hill-climber therefore moves *away* from the optimum. Deb and Goldberg establish that the necessary and sufficient condition for full deception is that all schemata of order $n-1$ favor the local optimum, which requires $z \ge \lfloor n/2 \rfloor$ [DG93].

2.2.4.6 Plateau

The Plateau function with parameter k is defined as OneMax everywhere except in a ball of radius k around the optimum, where fitness is held constant at the second-best value $n-k$ [AD21]:

$$\textsc{Plateau}_k(x) = \begin{cases} n-k & \text{if } n-k < \textsc{OneMax}(x) < n, \\ \textsc{OneMax}(x) & \text{otherwise.} \end{cases}$$

The global optimum is 1^n with value n. Unlike Jump, which penalizes solutions near the optimum with a *reduced* fitness, Plateau assigns the plateau region the same fitness as the boundary level $n-k$, so there are no true local optima. Instead, the algorithm performs an unguided random walk across the $\Theta(n^k)$ plateau points, gaining no fitness signal until it reaches 1^n by flipping between one and k zero-bits simultaneously.

Antipov and Doerr [AD21] establish that the expected runtime of the $(1+1)$ EA on $\textsc{Plateau}_k$ depends only on the probability of flipping between one and k bits per step, not on other properties of the mutation operator, and that the optimal mutation rate under standard bit mutation is approximately $k/(en)$, a factor of $\Theta(k)$ above the canonical $1/n$ recommendation. The full runtime expression is given in Appendix B.4.

2.2.4.7 Hierarchical If-and-Only-If (HIFF)

The Hierarchical If-and-Only-If (HIFF) function is a benchmark designed to model *hierarchical building-block interdependency*: a property absent from earlier Genetic Algorithms (GA) test problems, which treat sub-problems as separable [WHP98]. In HIFF, the optimal solution to a block of bits depends on how that block combines with its neighbors at the next level of the hierarchy, making sub-problems *non-separable* in a consistent, multi-level sense.

The fitness of a bitstring $x \in \{0,1\}^n$, where n must be a power of two, is defined recursively. A block B of length one always contributes a fitness of one. For any longer block, if all bits in B are identical (all zeros or all ones), the block contributes $|B|$ to fitness and its two halves are recursively evaluated; otherwise only the recursive evaluation of the two halves contributes. Formally,

$$f(B) = \begin{cases} 1 & \text{if } |B| = 1, \\ |B| + f(B_L) + f(B_R) & \text{if } |B| > 1 \text{ and all bits in } B \text{ are equal,} \\ f(B_L) + f(B_R) & \text{otherwise,} \end{cases}$$

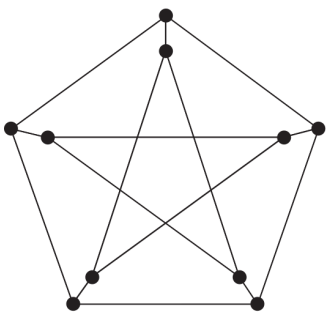
where B_L and B_R denote the left and right halves of B. The global optimum is achieved by the

all-zeros or all-ones bitstring, giving a maximum fitness of $n(\log_2 n+1)$. Two global optima exist and are maximally distant from one another, a structural feature that makes HIFF acutely challenging for hill-climbing algorithms: a single bit-flip from either optimum disrupts the top-level block uniformity and causes a sharp fitness drop, so local search stagnates at intermediate solutions in which only lower-level blocks are uniform [WHP98].

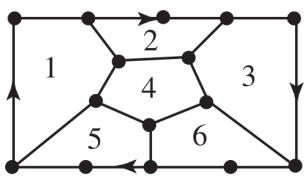
The landscape is highly multimodal. Secondary optima consist of strings with contiguous blocks of zeros and ones at each hierarchical level (for example, the first $n/2$ bits all zero and the remaining $n/2$ all one for a length-n string), and these local optima are maximally distant from both global optima. Oates et al. [OCL00] demonstrate empirically that this structure induces a characteristic tri-modal performance profile in evolutionary search: progress occurs in discrete phases corresponding to the successive exploitation of one-bit, two-bit, and higher-order mutations needed to escape each tier of local optima. This property makes HIFF a useful diagnostic benchmark for regret analysis, as the hierarchical barrier structure should produce a distinctive staircase signature in the cumulative regret trajectory.

2.2.4.8 MaxSAT

Maximum Satisfiability problem (MaxSAT) is the optimization variant of the Boolean Satisfiability problem (SAT). A propositional formula in Conjunctive Normal Form (CNF) is a conjunction of *clauses*, each of which is a disjunction of *literals* (a Boolean variable or its negation). Given such a formula with m clauses over n Boolean variables, the goal is to find a truth assignment that maximizes the number of satisfied clauses. MaxSAT is NP-hard in general [HSS04] and serves as an important benchmark for combinatorial optimization algorithms.



Petersen graph



embedding on the projective plane

Petersen Graph 3-Coloring. The *Petersen graph* [Weind] is a well-known 3-regular graph on ten vertices and fifteen edges whose chromatic number is exactly three, meaning a valid 3-colouring exists and no 2-colouring does. This project encodes the 3-colouring problem as a MaxSAT instance over $n=30$ binary variables, with one variable per vertex-color pair: variable $x_{v,c}$ is 1 if vertex v is assigned color $c \in \{0,1,2\}$. Three families of soft clauses encode the coloring constraints: ten *at-least-one* clauses (each vertex must receive at least one color), thirty *at-most-one* clauses (each vertex must not hold two colors simultaneously), and forty-five *edge-conflict* clauses (adjacent vertices must not share a color), giving $m=85$ clauses in total. Because the Petersen graph is 3-colourable, all 85 clauses can be satisfied simultaneously, so the global optimum is $f^*=85$ and is genuinely achievable [HSS04]. This is the key advantage over random MaxSAT instances, whose declared optima are rarely reachable in practice.

2.2.4.9 Tunable Landscapes: NK-Landscapes

NK-landscapes [KL87] are a parametric family of pseudo-Boolean functions controlled by two integers: N (the number of variables) and K (the number of epistatic interactions per variable, $0 \leq K \leq N-1$). The fitness of a string $x \in \{0,1\}^N$ is defined as

$$f(x) = \frac{1}{N} \sum_{i=1}^{N} f_i\Big(x_i, x_{\pi(i,1)}, \ldots, x_{\pi(i,K)}\Big),$$

where each local contribution $f_i : \{0,1\}^{K+1} \to [0,1]$ is drawn independently and uniformly at random, and $\pi(i,\cdot)$ denotes the K epistatic neighbors of variable i.

When $K = 0$ the contributions are independent, producing a smooth, additively separable landscape analogous to OneMax. As K increases toward $N-1$, epistatic interactions intensify and the landscape becomes increasingly rugged, with the number of local optima growing in the limit. NK-landscapes thus provide a tunable model of ruggedness and are widely used to study algorithmic scalability and robustness across a spectrum of problem difficulties.

2.2.5 Landscape Properties

The geometric properties of f determine which tier of analytical information is both available and useful. Convexity and Lipschitz smoothness, the properties that underpin first- and second-order guarantees, are defined in Appendix B.2.

Ruggedness and Multimodality. A landscape is *rugged* when it contains many local optima separated by fitness valleys. In such landscapes the gradient (where it exists) is deceptive: improving moves at the local scale lead away from the global optimum, and no first-order structure can be exploited. Ruggedness arises naturally from *epistatic interactions*, where the contribution of one variable to the objective depends on the values of others, preventing any separable decomposition. As K increases in the NK-landscape model (Section 2.2.4.9), epistasis intensifies and the number of local optima grows, providing a tunable axis of landscape difficulty. Ruggedness is the primary motivation for stochastic search heuristics: randomized perturbation and acceptance of worsening moves are the principal tools for escaping local optima that would permanently trap deterministic hill-climbers.

2.2.6 Purpose of Randomized Search Heuristics

Devising tailored, exact algorithms for arbitrary BBO problems is, in general, intractable. Without structural information about the objective function, there is no principled basis on which to prefer one search strategy over another, and even when some structure is known, the cost of exploiting it may be prohibitive. This motivates turning to Randomized Search Heuristic (RSH)s and other metaheuristics[3] as general-purpose black-box methods.

2.2.6.1 No-Free-Lunch Theorems

This motivation is placed on theoretical footing by the *No-Free-Lunch (NFL) theorems* [WM02], which formalize the impossibility of a universally superior algorithm. The theorems state that, for

[3] *Metaheuristics* are search techniques that generalise heuristics observed in natural processes (e.g. EAs, Particle Swarm Optimization) [AAS18]

any two black-box optimization algorithms, their performance averaged uniformly over all possible objective functions on a finite domain is identical; any gain achieved on one subset of functions is offset by a corresponding loss on the complementary subset [WM02]. No single algorithm can therefore outperform all others across every problem class: the choice of algorithm implicitly encodes assumptions about problem structure, and those assumptions can always be violated.

For the pseudo-Boolean domain $\{0,1\}^n$, any RSH that performs strongly on benchmarks such as OneMax or LeadingOnes must underperform on some complementary class of functions. Complexity results such as $\mathbb{E}[\tau_{f^*}] = \Theta(n \log n)$ for the $(1+1)$ EA on OneMax [DJW02] are meaningful purely because OneMax is a natural representative of unimodal functions. The NFL theorems therefore make conditioning comparisons on a structured problem class a theoretical necessity not a practical convenience, which motivates the benchmark suite defined in Section 2.2.8.

2.2.7 Randomized Search Heuristic Algorithms

2.2.7.1 Evolutionary Algorithms (EA)

Evolutionary algorithms are a broad family of population-based, randomized search heuristics inspired by biological evolution. It belongs to the family of *metaphor-based* metaheuristics, as shown in Figure 2.1. In discrete optimization, the canonical paradigm maintains one or more *candidate solutions* (bit strings) and iterates two operators: *mutation*, which perturbs a solution stochastically, and *selection*, which retains solutions according to their objective value [DJW02; DN20]. The simplest instance of this paradigm, the $(1+1)$ EA, serves as the theoretical baseline throughout this project.

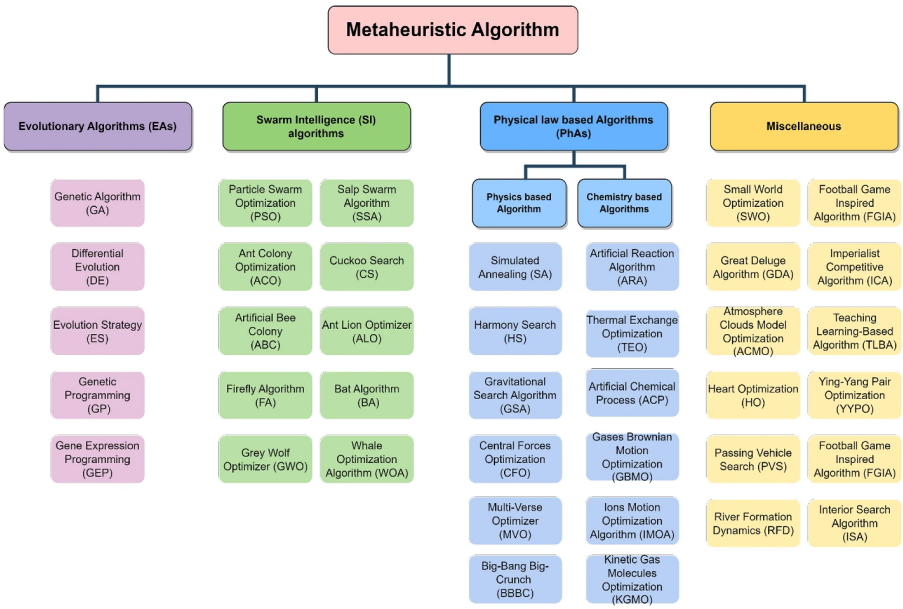


Figure 2.1: Taxonomy of Metaheuristics based on source of inspiration. EAs and SA are examples. Adopted from [RDD23, Figure 5]

The $(1+1)$ EA. The $(1+1)$ EA maintains a single current solution $x \in \{0,1\}^n$. In each iteration it produces one offspring y by *standard bit mutation*: every bit is flipped independently with probability p. The offspring replaces the parent if and only if $f(y) \geq f(x)$ (*elitist selection*). With the standard choice $p = 1/n$ the expected number of flipped bits is exactly one. The resulting process is a homogeneous Markov chain on $\{0,1\}^n$ with a non-decreasing fitness sequence [DJW02; Wit13].

Algorithm 1 The $(1+1)$ EA with mutation probability $p \in (0,1)$

```
Sample x ∈ {0,1}^n uniformly at random; evaluate f(x)
repeat
    y ← x
    for i = 1,...,n do
        flip y_i with probability p, independently
    end for
    if f(y) ≥ f(x) then x ← y
    end if
until termination criterion met
```

The classical runtime results for $p = 1/n$ are

$$\mathbb{E}[T_{\textsc{OneMax}}] = e\,n \ln n + O(n) \qquad \text{and} \qquad \mathbb{E}[T_{\textsc{LeadingOnes}}] = \tfrac{e-1}{2}\,n^2 + o(n^2),$$

and the $\Theta(n \log n)$ bound extends to *all* linear pseudo-Boolean functions with non-zero weights [Wit13]. On general landscapes the algorithm may require exponential time: even simple quadratic and unimodal functions can trap it indefinitely without restarts or adaptive mutation [DJW02]. Advanced variants and implementation-aware refinements are discussed in Appendix A.

2.2.7.2 Simulated Annealing

Algorithm 2 Simulated Annealing with cooling schedule $T(\cdot)$

```
Sample x ∈ {0,1}^n uniformly at random; evaluate f(x)
t ← 1;  x* ← x;  f* ← f(x)
repeat
    y ← x;  flip y_i for a uniformly random i ∈ {1,...,n}
    evaluate f(y);  δ ← f(y) − f(x)
    T ← max(T_min, T(t))                          ▷ clamp to avoid T = 0
    t ← t + 1
    if δ ≥ 0 or Uniform(0,1) < exp(δ/T) then
        x ← y;  f(x) ← f(y)                       ▷ Metropolis acceptance
    end if
    if f(x) > f* then x* ← x;  f* ← f(x)
    end if
until termination criterion met
return x*
```

Simulated Annealing (SA) is a probabilistic local search algorithm that extends iterative improvement by occasionally accepting worsening moves, allowing escape from local optima that would otherwise trap a pure hill-climber. The method was introduced by Kirkpatrick, Gelatt Jr, and Vecchi [KGV83], who drew an explicit analogy with the physical process of annealing in solids: a material is first melted at high temperature, then cooled slowly so that atoms settle into a stable, low-energy crystalline configuration.

In the optimization context, the algorithm maintains a current solution $x \in \mathcal{X}$ and at each step proposes a random neighbor $y \in \mathcal{N}(x)$. If $f(y) \geq f(x)$, the move is accepted unconditionally. If $f(y) < f(x)$, the move is accepted with probability

$$\exp\left(\frac{f(y) - f(x)}{T(t)}\right),$$

where $T(t) > 0$ is the *temperature* at step t, a control parameter that decreases over time according to a *cooling schedule* [KGV83; BT93]. At high temperatures the acceptance criterion is nearly uniform, permitting broad exploration; as $T(t) \to 0$ the algorithm collapses toward a greedy hill-climber that accepts only improvements.

The theoretical behavior of SA is governed by the *cooling schedule*. Modelling the process as a time-inhomogeneous Markov chain, Hajek [Haj88] established a sharp necessary and sufficient condition on the schedule for convergence in probability to the global optimum; the condition relates the decay rate of $T(t)$ to the depth d^* of the deepest non-global local minimum. The full statement and its implications are given in Appendix B.3. In practice, the schedule satisfying this condition cools logarithmically, which is theoretically optimal but often too slow; faster schedules reduce SA to a stochastic hill-climber once the temperature reaches zero. The choice of cooling schedule therefore encodes a fundamental trade-off between solution quality and computational budget, and is the primary design decision when applying SA to any combinatorial problem [KGV83; Haj88].

Cooling Schedules. Three cooling schedules were evaluated in this project.

Logarithmic cooling It decreases very slowly and is the only schedule considered here with a classical asymptotic convergence guarantee: for schedules of this form, convergence is obtained when $d \geq d^*$, d^* the depth (or height) of the largest local optima [Haj88].

Algorithm 3 Logarithmic cooling schedule

Require: scaling constant $d > 0$, evaluation index $t \geq 1$

1: **return** $T(t) = \dfrac{d}{\ln(t+1)}$ ▷ $d \geq d^*$ guarantees asymptotic convergence [Haj88]

Exponential cooling It follows the geometric pattern used in the original simulated annealing experiments [KGV83] and therefore reaches very low temperatures quickly under a finite evaluation budget.

Algorithm 4 Exponential cooling schedule

Require: initial temperature $T_0 > 0$, decay factor $\alpha \in (0,1)$, evaluation index $t \geq 0$

1: **return** $T(t) = T_0 \cdot \alpha^t$

Linear cooling It decreases uniformly to a prescribed floor and is used here as a practical fixed-horizon baseline in the experiments; unlike logarithmic cooling, it carries no classical asymptotic convergence guarantee in this setting [BT93].

Algorithm 5 Linear cooling schedule

Require: $T_0 > T_f \geq 0$, budget $t_{\max}$, evaluation index $t \geq 0$

1: **return** $T(t) = \max\Big(T_f,\, T_0 - (T_0 - T_f)\,\frac{t}{t_{\max}}\Big)$

Suite 11 (Subsection 5.4.8) compares the regret profiles of these schedules, focusing on when temperature collapse occurs and when instantaneous regret stabilizes.

2.2.7.3 Randomized Local Search (RLS)

Randomized Local Search (RLS) is a single-solution local search heuristic that at each step selects a neighbor of the current solution uniformly at random and moves to it if the objective value does not decrease (*elitist selection*). In the pseudo-Boolean setting, the standard neighborhood is the Hamming-1 ball: exactly one bit is chosen uniformly at random and flipped (‘local‘). [HS05b]

Three variants are studied in this project: greedy RLS, ε-greedy RLS, and decaying ε-greedy RLS.

Stochastic Hill-Climber - Greedy RLS The *greedy* RLS variant accepts only improving or equal-quality moves, making it a pure hill-climber with no mechanism to escape local optima.

Algorithm 6 Greedy RLS (Stochastic Hill Climber)

1: Sample $x \in \{0,1\}^n$ uniformly at random; evaluate $f(x)$
2: $t \leftarrow 1$; $x^* \leftarrow x$; $f^* \leftarrow f(x)$
3: **repeat**
4: $\quad y \leftarrow x$; flip y_i for a uniformly random $i \in \{1,\ldots,n\}$
5: $\quad$ evaluate $f(y)$; $t \leftarrow t+1$
6: $\quad$ **if** $f(y) \geq f(x)$ **then** $x \leftarrow y$; $f(x) \leftarrow f(y)$
7: $\quad$ **end if**
8: $\quad$ **if** $f(x) > f^*$ **then** $x^* \leftarrow x$; $f^* \leftarrow f(x)$
9: $\quad$ **end if**
10: **until** termination criterion met
11: **return** x^*

ε-greedy RLS The *ε-greedy* variant accepts a random Hamming-1 neighbor unconditionally with probability ε and otherwise applies elitist selection. This introduces a controlled exploration rate while still favoring improving moves most of the time.

Algorithm 7 ε-Greedy RLS

Require: exploration rate $\varepsilon \in (0,1)$

Sample $x \in \{0,1\}^n$ uniformly at random; evaluate $f(x)$
$t \leftarrow 1;\ x^* \leftarrow x;\ f^* \leftarrow f(x)$
repeat
  $y \leftarrow x;$ flip y_i for a uniformly random $i \in \{1,\dots,n\}$
  evaluate $f(y);\ t \leftarrow t+1$
  if Uniform$(0,1) < \varepsilon$ **then**
    $x \leftarrow y;\ f(x) \leftarrow f(y)$ ▷ explore unconditionally
  else if $f(y) \geq f(x)$ **then**
    $x \leftarrow y;\ f(x) \leftarrow f(y)$ ▷ exploit
  end if
  if $f(x) > f^*$ **then** $x^* \leftarrow x;\ f^* \leftarrow f(x)$
  end if
until termination criterion met
return x^*

Decaying ε-greedy RLS The *decaying ε-greedy* variant reduces the exploration probability over time according to $\varepsilon_t = \varepsilon_0/t$. This concentrates exploration early in the search budget and gradually converges toward greedy behavior as the run progresses.

Algorithm 8 Decaying-ε-Greedy RLS

Require: initial exploration rate $\varepsilon_0 \in (0,1)$

Sample $x \in \{0,1\}^n$ uniformly at random; evaluate $f(x)$
$t \leftarrow 1;\ x^* \leftarrow x;\ f^* \leftarrow f(x)$
repeat
  $y \leftarrow x;$ flip y_i for a uniformly random $i \in \{1,\dots,n\}$
  evaluate $f(y);\ \varepsilon_t \leftarrow \varepsilon_0/t;\ t \leftarrow t+1$
  if Uniform$(0,1) < \varepsilon_t$ **then**
    $x \leftarrow y;\ f(x) \leftarrow f(y)$ ▷ explore (probability decays as $1/t$)
  else if $f(y) \geq f(x)$ **then**
    $x \leftarrow y;\ f(x) \leftarrow f(y)$ ▷ exploit
  end if
  if $f(x) > f^*$ **then** $x^* \leftarrow x;\ f^* \leftarrow f(x)$
  end if
until termination criterion met
return x^*

2.2.8 Performance Measures in BBO

The theoretical analysis of BBO algorithms rests on three complementary frameworks, each fixing one dimension of the problem and analyzing what can be guaranteed along the other.

Expected runtime. The canonical measure is the *expected optimization time*: the expected number of evaluations until an optimal solution is first queried [DJW02; DN20]. Formally, the *first hitting*

time of the global optimum is

$$\tau_{f^*} := \min\{t \in \mathbb{N} : f(x_t) = f^*\},$$

and performance is summarized by $\mathbb{E}[\tau_{f^*}]$. Canonical results are $\Theta(n \log n)$ for the $(1+1)$ EA on OneMax, $\Theta(n^2)$ on LeadingOnes, and $\Theta(n^k)$ on Jump_k [DJW02], derived via drift theory [Len19] and Markov chain arguments [DN20].

Fixed-budget analysis. In *fixed-budget analysis* [JZ14] the budget T is fixed and the quantity of interest is $\mathbb{E}[f_T^*(\mathcal{A})]$, the expected quality of the best solution found within T evaluations. This is practically motivated: a user typically cannot detect optimality and the run terminates at budget exhaustion regardless. Budget-based rankings can reverse those from $\mathbb{E}[\tau_{f^*}]$, and maximizing $\mathbb{E}[f_T^*]$ is equivalent to minimizing expected simple regret at budget T (Section 2.4.2.3).

Fixed-Target Analysis and Runtime Profiles. *Fixed-target analysis* extends the single hitting time to every intermediate fitness level v, characterizing the distribution of

$$\tau_v := \min\{t \in \mathbb{N} : f(x_t) \geq v\}.$$

Pinto and Doerr [PD18] show that reporting the full collection $(\mathbb{E}[\tau_v])_v$ reveals crossover fitness levels where one algorithm overtakes another, behavior invisible to $\mathbb{E}[\tau_{f^*}]$ alone. Chapter 3 develops this further.

Shared limitation. All three frameworks measure performance at isolated thresholds rather than over the entire quality trajectory, and none penalizes an algorithm for spending many evaluations at low fitness. A strategy that stagnates and then finds the optimum in a single lucky step is indistinguishable from one making steady progress under $\mathbb{E}[\tau_{f^*}]$ alone. Pinto and Doerr [PD18] call explicitly for *combined anytime measures* from machine learning, naming regret as a natural candidate while noting that a satisfying formulation remains open.

2.3 Sequential and Online Optimization Frameworks

The regret framework adopted in this project originates in the online learning and sequential decision-making literature. This section introduces the minimal concepts from that tradition, the exploration–exploitation trade-off, the multi-armed bandit model, and Bayesian optimization, that are required to make the regret vocabulary of Section 2.4 rigorous and to situate this project within the broader landscape of black-box optimization research.

2.3.1 Exploration-Exploitation Trade-off

Sequential black-box optimization is governed by the exploration–exploitation trade-off. At each round, the algorithm must either explore uncertain regions of the search space to gain information or exploit the currently best-known region to obtain good values immediately [CL06; BC12; LS20] Since each evaluation is both a measurement and a search step, neither extreme is satisfactory: pure exploitation can lock onto a suboptimal region, while pure exploration wastes budget.

This tension is central to regret-based analysis. Regret measures the cost of not knowing the objective in advance, so it captures both slow identification of good regions and repeated evaluation

of poor ones.

2.3.2 Multi-Armed Bandits

The Multi-Armed Bandits (MAB) model is the basic framework for sequential decision-making under partial feedback [BC12; LS20]. In the stochastic K-armed setting, an algorithm chooses an arm $A_t \in \{1, \ldots, K\}$ at each round and observes only the reward of that arm. If arm i has mean reward μ_i and $\mu^* = \max_i \mu_i$, the standard pseudo-regret is

$$\bar{R}_T = T\mu^* - \sum_{t=1}^{T} \mu_{A_t} = \sum_{i:\Delta_i>0} \Delta_i \, \mathbb{E}[N_i(T)],$$

where $\Delta_i = \mu^* - \mu_i$ is the sub-optimality gap and $N_i(T)$ is the number of pulls of arm i up to time T [BC12; LS20]. Regret is therefore driven by how often the learner samples suboptimal actions.

2.3.3 Bayesian Optimization

The closest existing body of work applying regret to black-box optimization is Bayesian Optimization (BO) [JSW98; Sha+15; Fra18]. BO treats the optimization as a sequential learning problem: a probabilistic surrogate (typically a Gaussian process) models the unknown objective, and an acquisition function selects the next query by trading predicted value against uncertainty. Incidentally, it has yielded a mature regret theory: the GP-UCB algorithm admits sublinear cumulative-regret guarantees under Gaussian-process or RKHS[4] assumptions [Sri+09; CG17; SBC17]. These guarantees rely on surrogate assumptions that are natural in low-dimensional continuous spaces but do not transfer directly to the pseudo-Boolean domains $\{0,1\}^n$ studied here, where kernel engineering is required and acquisition optimization is itself an NP-hard combinatorial problem [Sha+15]. This motivates studying regret for the simpler, surrogate-free randomized search heuristics that form the subject of this project.

2.4 Computational Regret

Having established both the black-box setting and the online learning framework from which regret originates, this section defines regret formally and identifies the specific variant, incumbent cumulative regret, that is used as the primary performance measure throughout the experimental analysis in Chapter 5.

2.4.1 Definition of Regret

Let $f : \mathcal{X} \to \mathbb{R}$ be an objective function with global optimum value $f^* = \max_{x \in \mathcal{X}} f(x)$. An algorithm that queries points $x_1, x_2, \ldots, x_T$ incurs *regret* by querying suboptimal solutions. Regret quantifies the gap between the algorithm's performance and that of an oracle always querying the optimum [CL06].

[4] *abbr.* Reproducing kernel Hilbert space. (Optionally, see [Gho+21])

2.4.2 Types of Regret

2.4.2.1 Instantaneous Regret

The *instantaneous regret* at round t is

$$\rho_t = f^* - f(x_t),$$

capturing the suboptimality of the single query made at step t. A related variant tracks the *incumbent* (best-so-far) solution rather than the current query:

$$\rho_t^{\text{inc}} = f^* - f(\hat{x}_t), \quad \hat{x}_t = \arg\max_{s \le t} f(x_s).$$

Because $\hat{x}_t$ is non-decreasing in quality, ρ_t^{inc} is monotonically non-increasing over t, and is equivalent to simple regret evaluated at time t(2.4.2.3).

2.4.2.2 Cumulative Regret

Two cumulative regret variants arise naturally and serve different purposes.

The *Cumulative Regret (CR)* after T rounds is [CL06]

$$R_T = \sum_{t=1}^{T} \rho_t = \sum_{t=1}^{T} (f^* - f(x_t)).$$

This is the standard formulation from the online learning literature: it sums the instantaneous suboptimality of each individual query, regardless of what the algorithm has previously observed. Minimizing R_T corresponds to performing well *throughout* the optimization process, penalizing every suboptimal query equally.

The *incumbent cumulative regret* instead tracks the best-so-far process

$$B(t) := \max_{s \le t} f(x_s),$$

yielding

$$R_T^{\text{inc}} = \sum_{t=1}^{T} \rho_t^{\text{inc}} = \sum_{t=1}^{T} (f^* - B(t)).$$

Since $B(t)$ is non-decreasing in t, the per-step contributions ρ_t^{inc} are non-increasing, and $R_T^{\text{inc}} \le R_T$ for all T. The incumbent formulation is the appropriate notion when the practitioner retains the best solution found rather than only the most recent query, which is the standard convention for the randomized search heuristics studied in this project. All regret analyses in Chapter 5 use R_T^{inc} unless otherwise stated.

In continuous time, the incumbent cumulative regret is approximated via a left-hold (piecewise-constant) integral

$$R_T^{\text{inc}} \approx \sum_{t=1}^{T} \rho_{t-1}^{\text{inc}} \Delta t_t, \quad \Delta t_t = t - t_{t-1},$$

which accounts for unequal spacing between evaluations when function evaluation time is non-uniform. In all experiments reported here, evaluations are unit-cost and time is measured in discrete query counts, so $\Delta t_t = 1$ throughout and this reduces to the standard discrete sum.

2.4.2.3 Simple Regret

The *Simple Regret (SR)* after T rounds is [BMS11]

$$r_T = f^* - f(\hat{x}_T),$$

where $\hat{x}_T = \arg\max_{t \le T} f(x_t)$ is the best solution encountered. Simple regret measures the quality of the *final output* and is the appropriate notion when the goal is to return a good solution rather than to perform well during the search. Note that $r_T = \rho_T^{\text{inc}}$: simple regret is the incumbent instantaneous regret at the final round.

When comparing algorithms across multiple independent runs, the standard scalar summary is the *expected simple regret*

$$\mathbb{E}[r_T] = \frac{1}{N} \sum_{i=1}^{N} r_T^{(i)},$$

where $r_T^{(i)}$ denotes the simple regret of the i-th run.

2.4.2.4 Normalized Regret

To enable comparison across problem instances with different objective scales, simple regret can be normalized by the range of attainable fitness values:

$$\tilde{r}_T = \frac{f^* - f(\hat{x}_T)}{f^* - f_{\text{worst}}} \in [0, 1],$$

where f_{worst} is the worst observed or theoretical fitness value. A value of 0 indicates an optimal solution was found; a value of 1 indicates no improvement over the worst case. When $f^* = f_{\text{worst}}$ (a degenerate constant landscape), the problem is trivially solved and normalized regret is defined as 0.

2.4.3 Regret Bounds

The central benchmark of regret analysis is *sublinear regret*: if $R_T = o(T)$, then the average per-round regret $R_T/T \to 0$, meaning the algorithm asymptotically performs as well as an oracle with full information [CL06; BC12; LS20]. Classic results in finite stochastic bandits establish logarithmic gap-dependent upper bounds and matching lower bounds, showing that logarithmic regret is unavoidable up to constants under standard assumptions [ACF02; LR85]. In Gaussian-process bandits, upper and lower bounds depend additionally on the information-gain structure of the kernel [Sri+09; SBC17]. These results are important here primarily as a template: regret provides a well-grounded complexity measure for sequential optimization, and sublinear regret is the performance target against which any algorithm is judged. The per-algorithm bounds are discussed in the context of related work in Section D.0.1.

2.5 Regret-Aware Black-Box Optimization

Regret provides a fine-grained complexity measure in the black-box setting: rather than simply asking whether an algorithm finds the optimum within T queries, regret tracks how suboptimal the queries are throughout the process. This is particularly relevant when every evaluation is costly and performing well before the budget is exhausted matters in practice.

2.5.1 Regret as a Performance Measure for Randomized Search Heuristics

For randomized search heuristics, regret complements rather than replaces runtime analysis. Runtime asks when a target is first reached; regret asks how good the algorithm's queried or retained solutions are throughout the run. Fixed-budget analysis already moves in this direction, since expected simple regret is just the expected final sub-optimality after a budget of T evaluations [JZ14; BMS11].

The distinction matters operationally. Low simple regret is appropriate when only the final returned solution matters. Low cumulative regret is appropriate in anytime settings, where useful intermediate performance also matters [CL06; PD18]. For the heuristics considered in this project, the incumbent-based formulation is the natural one because the algorithm's output at interruption is the best-so-far solution.

Chapter 3

Runtime Profiles and Anytime Performance

3.1 Motivation

Section 2.2.8 surveyed the three standard frameworks for measuring the performance of black-box optimization algorithms: the expected runtime $\mathbb{E}[\tau_{f^*}]$, fixed-budget analysis, and fixed-target analysis culminating in the runtime profile framework from [PD18]. Each improves on its predecessor in descriptive richness, yet none provides a single scalar that captures solution quality at every point in the run. Fixed-budget analysis, for instance, compresses the full trajectory to one value per budget [JZ14]. What the practitioner actually observes is the evolution of solution quality over the *entire* course of the run.

Pinto and Doerr make the case for this richer mode of reporting directly:

> Our intention is rather to highlight to researchers working on theoretical aspects in evolutionary computation that, beyond being possibly more relevant for practitioners, explicitly reporting runtime profiles can give quite interesting insights into the performance of EAs. ([PD18], p. 19)

This chapter takes that framing as its starting point. Building on the notation established in Section 2.2.8, this chapter formalizes the distributional runtime profile as a mathematical object, and then proves that expected cumulative regret decomposes exactly as a double sum over this profile. The result unifies the runtime-analysis and regret-based perspectives into a single identity, providing the *combined anytime measure* Pinto and Doerr [PD18] call for.

3.2 Variants

3.2.1 Expectation-Based Runtime Profiles

Building on the first hitting time τ_v defined in Section 2.2.8, Pinto and Doerr [PD18] propose reporting the full collection of expected first hitting times $\left(\mathbb{E}[\tau_v]\right)_v$ across all relevant target levels, the *expectation-based runtime profile*. When canonical fitness levels exist, such as for OneMax or LeadingOnes, these provide natural targets; for other functions, intermediate values identified during analysis or by linear interpolation of the fitness range may be used [PD18]. No new algorithms or

proof techniques are required: the profile is typically already implicit in existing runtime proofs and calls only for a richer mode of presentation.

The practical value of this richer view is illustrated by the behavior of $(1+1)$ EA variants on LeadingOnes: reporting only the expected optimization time suggests RLS is superior, yet the runtime profile reveals that $(1+1)$ $\text{EA}_{>0}$ (*App. para.* A.2) reaches all intermediate fitness levels $i \leq 7{,}980$ faster than RLS for problem size $n = 10{,}000$, with RLS only pulling ahead in the final portion of the run [PD18]. Such crossover behavior is directly relevant to the design of adaptive parameter and operator selection schemes.

3.2.2 Distributional Runtime Profiles

The expectation-based runtime profile of [PD18] summarizes the distribution of τ_v by its mean. A finer characterization, and the appropriate quantity for connecting runtime profiles to regret-based measures, is the *distributional runtime profile*

$$\mathcal{P}(v,t) := \mathbb{P}(\tau_v > t), \tag{3.1}$$

which records, for each target level v and time step t, the probability that level v has not yet been reached by time t. The expectation-based profile is recovered by the standard tail-sum formula for non-negative integer-valued random variables [DN20, p. 16]:

$$\mathbb{E}[\tau_v] = \sum_{t=0}^{\infty} \mathbb{P}(\tau_v > t). \tag{3.2}$$

The distributional profile is the more primitive object from which expectation-based statements follow, and it constitutes the formal link between runtime profiles and the cumulative regret framework developed below.

Remark 2 (Naming convention)**.** *The term* runtime profile *appears in two distinct senses in the literature and in this work. [PD18] uses it to denote the* expectation-based *runtime profile* $(\mathbb{E}[\tau_v])_v$*, which summarizes per-level performance by expected hitting times. The present work additionally employs the strictly finer* distributional *runtime profile* $\mathcal{P}(v,t) = \mathbb{P}(\tau_v > t)$*, from which the expectation-based profile is recovered via the tail-sum formula (Equation 3.2). To avoid ambiguity, both notions are referred to explicitly whenever both are in scope. Hereafter, and in all practical experimentation in this report,* runtime profile *refers to the distributional variant* $\mathcal{P}(v,t)$ *unless otherwise stated.*

3.3 Connection to Cumulative Regret

One of the directions suggested by Pinto and Doerr in their *Future Works* section is restated for emphasis:

> We feel that there is a need for combined measures that are capable of describing the anytime behavior of an EA. Regret-based measure as used in machine learning could be a key here, but we haven't been able so far to identify a fully satisfying measure. ([PD18], p. 22)

As far as we know, establishing such a relationship remains an open question. This section consolidates the following proposal: that the *Cumulative Regret (CR)* (Section 2.4.2.2) and the distributional

runtime profile are not independent approaches to algorithm performance but are two representations of the same underlying quantity. The identity below attempts to establish this relation.

The relationship can be made explicit through a formal setup. A randomized black-box algorithm produces an adaptive sequence of query points $x_1, x_2, \ldots \in \{0,1\}^n$. Recall from Section 2.4.2.2 the *best-so-far process*

$$B(t) := \max_{s \le t} f(x_s),$$

which records the best objective value seen up to time t. Since $B(t)$ is non-decreasing in t, the event $\{B(t) < v\}$ is equivalent to $\{\tau_v > t\}$: the level v has not yet been hit. The incumbent cumulative regret over a budget of T evaluations is therefore

$$\mathrm{CR}(T) := \sum_{t=1}^{T} (f^* - B(t)).$$

The proof of the main theorem relies on two elementary lemmas.

Lemma 1 (Tail-sum representation for the gap)**.** *For any integer b with $0 \le b \le f^*$,*

$$f^* - b = \sum_{v=1}^{f^*} \mathbf{1}[b < v].$$

Proof. The right-hand side counts the integers $v \in \{1, \ldots, f^*\}$ that strictly exceed b, namely the set $\{b+1, \ldots, f^*\}$, which has cardinality $f^* - b$. □

Lemma 2 (Indicator equivalence)**.** *For all $v \in \{1, \ldots, f^*\}$ and $t \in \mathbb{N}$,*

$$\mathbf{1}[B(t) < v] = \mathbf{1}[\tau_v > t].$$

Proof. Since $B(t)$ is non-decreasing in t, the event $\{B(t) \ge v\}$ is equivalent to $\{\tau_v \le t\}$. Taking complements gives $\{B(t) < v\} = \{\tau_v > t\}$, so the two indicators agree on every sample path. □

The following identity connects *expected* cumulative regret to the distributional runtime profile.

Theorem 3 (Tail-Sum Regret Representation)**.** *For any randomized black-box algorithm operating on f with finite budget $T \in \mathbb{N}$,*

$$\mathbb{E}[\mathrm{CR}(T)] = \sum_{v=1}^{f^*} \sum_{t=1}^{T} \mathbb{P}(\tau_v > t).$$

Proof. For each t, Lemma 1 applied to $b = B(t)$ gives

$$f^* - B(t) = \sum_{v=1}^{f^*} \mathbf{1}[B(t) < v].$$

Summing over $t = 1, \ldots, T$ yields

$$\mathrm{CR}(T) = \sum_{t=1}^{T} (f^* - B(t)) = \sum_{t=1}^{T} \sum_{v=1}^{f^*} \mathbf{1}[B(t) < v].$$

By Lemma 2, $\mathbf{1}[B(t) < v] = \mathbf{1}[\tau_v > t]$ on every sample path, so

$$\mathrm{CR}(T) \;=\; \sum_{t=1}^{T}\sum_{v=1}^{f^*} \mathbf{1}[\tau_v > t].$$

Taking expectations and applying linearity of expectation together with $\mathbb{E}[\mathbf{1}[\tau_v > t]] = \mathbb{P}(\tau_v > t)$ gives

$$\mathbb{E}[\mathrm{CR}(T)] \;=\; \sum_{t=1}^{T}\sum_{v=1}^{f^*} \mathbb{P}(\tau_v > t).$$

Since both sums are finite, their order may be exchanged, yielding the stated result. □

Theorem 3 expresses expected cumulative regret as a double sum of the distributional runtime profile: $\mathbb{E}[\mathrm{CR}(T)]$ is the total expected time the algorithm spends below each fitness level v, summed over all levels. An algorithm with a uniformly smaller runtime profile (one that reaches each fitness level earlier in expectation) therefore incurs lower cumulative regret. The regret measure thus simultaneously rewards early progress at every tier of the landscape, formalizing the intuition that steady improvement throughout a run is preferable to stagnation followed by a late breakthrough, even when both strategies share the same expected optimization time.

Taking the infinite-horizon limit, and applying the tail-sum identity for $\mathbb{E}[\tau_v]$ along with the monotone convergence theorem, yields the following asymptotic connection.

Corollary 4 (Asymptotic Regret and Expected Hitting Times)**.** *Suppose that τ_v is almost surely finite with finite expectation for each $v \in \{1, \ldots, f^*\}$. Then*

$$\lim_{T\to\infty} \mathbb{E}[\mathrm{CR}(T)] \;=\; \sum_{v=1}^{f^*} \mathbb{E}[\tau_v] \;-\; f^*.$$

Proof. **Step 1: Tail-sum identity for $\mathbb{E}[\tau_v]$.** Since $\tau_v \in \mathbb{N}_{\geq 1}$ by the convention that $t = 1$ is the first oracle call and initialization is excluded from the query budget, we have $\mathbb{P}(\tau_v > 0) = 1$. The standard tail-sum formula for a positive integer-valued random variable then gives

$$\mathbb{E}[\tau_v] \;=\; \sum_{t=0}^{\infty} \mathbb{P}(\tau_v > t) \;=\; 1 + \sum_{t=1}^{\infty} \mathbb{P}(\tau_v > t),$$

hence $\sum_{t=1}^{\infty} \mathbb{P}(\tau_v > t) = \mathbb{E}[\tau_v] - 1$.

Step 2: Passing to the limit in Theorem 3. For each fixed v, the partial sums $\sum_{t=1}^{T} \mathbb{P}(\tau_v > t)$ are non-decreasing in T and non-negative, so the monotone convergence theorem gives

$$\lim_{T\to\infty} \sum_{t=1}^{T} \mathbb{P}(\tau_v > t) \;=\; \sum_{t=1}^{\infty} \mathbb{P}(\tau_v > t) \;=\; \mathbb{E}[\tau_v] - 1.$$

Step 3: Summing over v. Applying Step 2 to each $v = 1, \ldots, f^*$ and invoking Theorem 3,

$$\lim_{T\to\infty} \mathbb{E}[\mathrm{CR}(T)] \;=\; \sum_{v=1}^{f^*} \big(\mathbb{E}[\tau_v] - 1\big) \;=\; \sum_{v=1}^{f^*} \mathbb{E}[\tau_v] \;-\; f^*.$$ □

Corollary 4 provides a unifying perspective: the long-run expected cumulative regret decomposes as the sum of the expected first hitting times over all fitness levels, minus the constant f^* arising from

the $t = 0$ term in the tail-sum identity. Algorithms that reach all intermediate fitness levels quickly are therefore those that incur low long-run cumulative regret. This formalizes the intuition advanced in [PD18] that an algorithm making steady progress through the fitness landscape is preferable to one that stagnates at low fitness values even if both ultimately find the optimum in the same expected time.

The identity also reveals that the regret-based measures studied in the online learning literature [CL06] and the runtime profile framework of [PD18] are not independent approaches to measuring algorithm performance but are two views of the same underlying quantity, connected through the distributional runtime profile $\mathcal{P}(v,t)$.

3.4 Practical Relevance

As discussed in Section 2.2.8, the limitations of classical performance measures stem from their treatment of optimization as a sequence of threshold events rather than as a continuous quality trajectory.

Pinto and Doerr [PD18] note that runtime profiles are particularly valuable for the design of non-static parameter and operator selection strategies in EAs (such as adaptive mutation rates, crossover probabilities, and selection pressure) since such strategies must make decisions based on current solution quality rather than on the expected optimization time alone. The identification of *crossover fitness levels* (where one algorithm overtakes another in the runtime profile) provides actionable guidance for when to switch between strategies.

Furthermore, as established in Theorem 3, any collection of upper bounds on $\mathbb{E}[\tau_v]$ for all levels v simultaneously yields an upper bound on expected cumulative regret, and conversely, lower bounds on regret propagate to lower bounds on a weighted aggregate of first hitting times. Cumulative regret thus provides a single scalar summary of the entire runtime profile, complementing the per-level view that the profile itself offers. The two representations (regret as a scalar, runtime profile as a function) are therefore best understood as complementary instruments for the same measurement task, connected by the identity of Theorem 3.

Chapter 4

Supplementary Software Design

This chapter discusses the supplementary software developed for carrying out multiple experiments on *simulated* BBO problems efficiently. Trivially, analysis cannot be carried out on truly black-box instances; the problems are emulated to be black-box by providing only the *zeroth-order* information.

4.1 Purpose of the Software

A supplementary experimental framework, `regret`, was developed to support the orchestration and analysis of experiments on pseudo-Boolean and combinatorial optimization problems. The framework was motivated by the absence of readily available tooling that combines (i) flexible, declarative batch experiment execution, (ii) trajectory-level data collection, and (iii) regret-specific analysis pipelines in a single cohesive system.

Existing platforms such as Google Vizier [Gol+17] and EvoX [Hua+24] address hyperparameter optimization and population-based search respectively, but neither provides native support for regret metrics, runtime profile analysis, or the configuration-first experiment execution model required for this project's experimental protocol. `regret` was therefore purpose-built to fill this gap, providing a reproducible, extensible environment for studying simple, instantaneous, and cumulative regret across a controlled suite of algorithms and benchmark problems.

4.1.1 Preexisting software

Several established frameworks were considered prior to development. **Google Vizier** [Gol+17] provides a scalable black-box optimization service with a client–server architecture, but is oriented toward hyperparameter tuning rather than controlled algorithmic benchmarking, and does not expose trajectory-level data in a form amenable to regret analysis. **EvoX** [Hua+24] is a JAX-based evolutionary computation library offering GPU-accelerated population methods, but its abstraction layer couples algorithm state to framework internals in ways that preclude the lightweight callback-driven trajectory recording required here. General-purpose scientific workflow systems such as Snakemake [Möl+25] and Luigi [Eri12] offer pipeline orchestration but impose significant configuration overhead for the combinatorial (algorithm $\times$ problem $\times$ budget) sweeps central to this study.

4.1.2 Capabilities

The framework supports the full experimental life-cycle through a single Command-Line Interface (CLI), `run_experiment`, which accepts one or more YAML configuration files and executes the appropriate sub-command. Table 4.1 summarizes the available commands.

Command	Description
`validate`	Parse and validate one or more YAML configurations against the JSON Schema and semantic registry checks without executing any computation.
`plan`	Print a dry-run summary: problem/algorithm combinations, total runs, and evaluation budgets.
`run [--no-plot]`	Execute one or more experiment configurations; optionally suppress plot generation after completion.
`analyze`	Regenerate all plots from previously saved JSON results without re-running experiments.
`table [--format]`	Generate summary tables from existing results in `latex`, `csv`, or `markdown` format.

Table 4.1: CLI sub-commands exposed by the `run_experiment` entry point (equivalently `python -m regret`).

Batch Experiment Execution. All experiments are specified declaratively via YAML configuration files validated against a JSON Schema prior to execution. A configuration defines the suite name, evaluation budgets, run count, output mode, and lists of problem and algorithm specifications. Problems and algorithms are referenced by string keys resolved against internal registries, providing a clean separation between configuration and implementation.

Parallelized execution. When `parallel: true` is set in the YAML configuration, independent runs are initiated via Python's `ProcessPoolExecutor`, with per-run random seeds fixed to indices $0, 1, \ldots, r-1$ to ensure full reproducibility. Results are reordered by seed index after completion.

Flexible output modes. In `lite` mode, only summary statistics are persisted per run. In `full` mode, the complete evaluation trajectory (a sequence of `(evaluation_index, current_value, best_value)` tuples) is recorded and stored alongside aggregate statistics in structured JSON files.

Batched multi-configuration execution. All sub-commands accept multiple configuration files, enabling sequential execution of logically grouped experiment suites from a single invocation.

4.1.3 Design Decisions

The software was intended to be extensive and configurable for other developers and users to interact with quickly, with options for modularity when required. This was enabled by the adoption of configuration-first, job dispatch policy for independent and declarative experimentation. Problems and Algorithms to test against can easily be defined by extending abstract classes `Problem` and `Algorithm`, ensuring a common interface for consistent interactions.

4.1.3.1 Independent Job Execution framework

The CLI is implemented using **Typer** [Ram19], which maps sub-commands to typed Python functions with automatic argument parsing. The "orchestration" layer coordinates problem and algorithm instantiation, result persistence, and plot generation, delegating individual runs to `ExperimentRunner`. Parallel execution

uses `ProcessPoolExecutor` from Python's standard `concurrent.futures` module; results are collected via `as_completed` and reordered by seed for deterministic output ordering. Figure 4.1 illustrates the full validation and the experiment (job) dispatch pipeline.

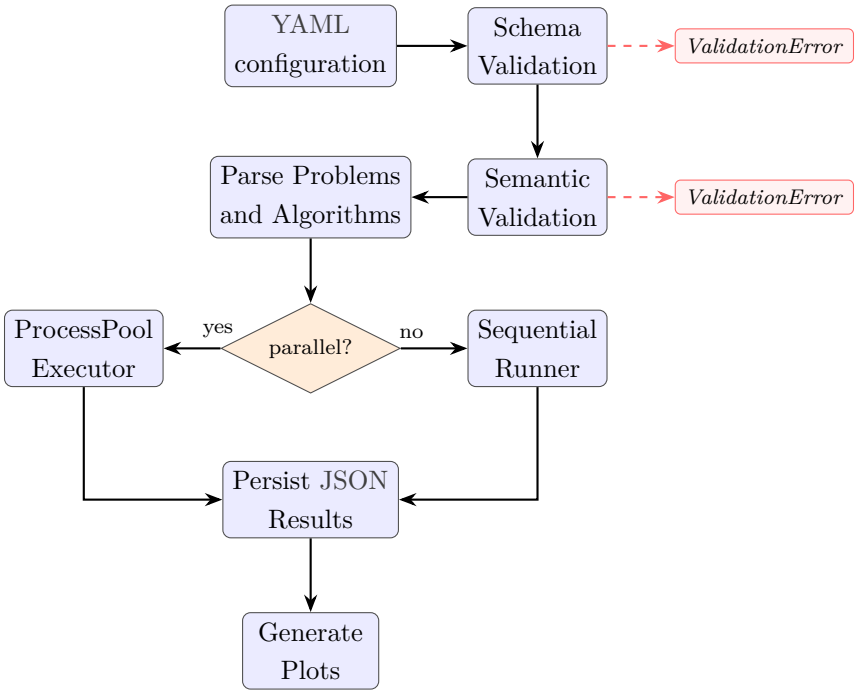


Figure 4.1: Validation and job dispatch pipeline. YAML configuration files are validated structurally via JSON Schema and then semantically against class registries before execution begins. Any validation failure raises a `ValidationError` and halts the pipeline before computation begins.

4.1.3.2 Raw Output JSON files

The output JSON files are *compacted* by replacing redundant stylistic whitespaces and newlines with other separators. This reduced the raw file sizes by up to 50%. JSON files were preferred for storing the raw files since it included *mixed* data, including metadata, general statistics for each run, along with the trajectories. A 'NoSQL' (key-value) database was considered but not used since the experiments were relatively small scale.

4.1.3.3 UML Diagram

Figure 4.2 presents the core class structure of the framework. Abstract base classes `Problem` and `Algorithm` define the shared interface enforced across all concrete implementations. Concrete subclasses are registered by string key and resolved at runtime. `ExperimentRunner` orchestrates individual runs and delegates result persistence, while the `CoolingSchedule` protocol is resolved from YAML at parse time via a *registry*.

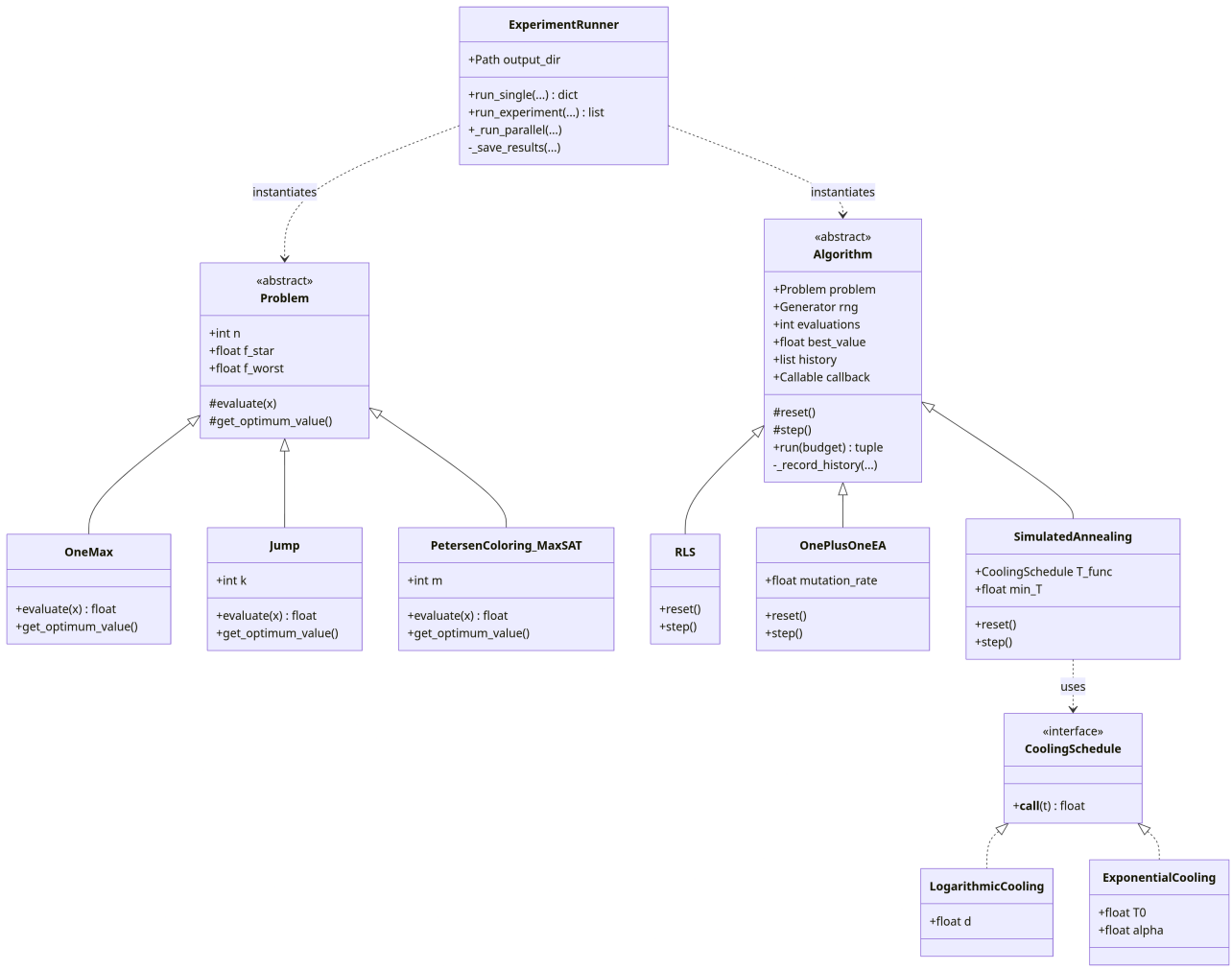


Figure 4.2: Simplified UML class diagram of the `regret` framework. Solid triangle arrows (▷) denote inheritance; dashed arrows denote dependency or instantiation. Hashes '#' indicate abstract declarations. Several concrete subclasses (e.g. `MuPlusLambdaEA`, `RLSExploration`, `NKLandscape`, `LinearCooling`) are omitted for clarity.

4.1.4 Limitations

The framework is currently constrained to **binary search spaces** $\{0,1\}^n$; continuous and integer-valued domains are not supported. **Multi-objective optimization** is not implemented. The NK landscape optimum computation uses exhaustive search, restricting its practical use to $n \leq 20$. Runtime profile analysis assumes integer-valued, unit-increment fitness functions for the exact tail-sum identity:

$$\mathbb{E}[\mathrm{CR}(T)] \;=\; \sum_{v=1}^{f^*} \sum_{t'=1}^{T} \big[1 - P(\tau_v \leq t')\big];$$

continuous fitness functions (e.g.`NK-Landscape`) yield approximate results and may exhibit minor discrepancies in the $\mathbb{E}[\mathrm{CR}(T)]$ verification plot.

4.1.5 Result Interpretations

All experiment output is written under a structured directory hierarchy. Raw JSON result files are stored at:

```
results/raw/<suite>/<problem>/<algorithm>/n<N>/b<B>.json
```

where `<suite>`, `<problem>`, and `<algorithm>` are normalized display names, `N` is the problem dimension, and `B` is the evaluation budget. Each file contains three top-level keys: `metadata` (algorithm, problem class, dimension,

budget, run count, mode, and timestamp), `statistics` (mean, median, standard deviation, quartiles, and probability of optimum across all runs), and `results` (an ordered list of per-run dictionaries each containing regret, best value, optimality flag, evaluation count, seed, and the complete trajectory (in `full` mode)).

Generated figures are written under `results/figures/` following an analogous hierarchy:

`results/figures/<suite>/<problem>/n<N>/<subdir>/<filename>.pdf`

Plot subdirectories are configurable via `layout.structure` in the YAML configuration; the defaults are `aggregate/` for cross-budget plots, `history/` for trajectory and regret time-series plots, and `distributions/` for budget-specific box-plots and performance profiles. Runtime profile plots are written to a `profiles/` subdirectory within the raw results tree. All figures are saved as PDF at 300 DPI. Table 4.2 summarizes the full output organization.

Path pattern	**Contents**
Raw data (JSON)	
`results/raw/` `<suite>/<prob>/<algo>/n<N>/b<B>.json`	Per-budget result file: metadata, aggregate statistics, per-run regrets, and (in `full` mode) full evaluation trajectories.
`results/raw/<suite>/<prob>/` `profiles/n<N>/cr_profile_data_<algo>.csv`	Runtime profile CSV; columns: `evaluation`, `expected_cumulative_regret`, `mean_cumulative_regret`.
Figures (PDF, 300 DPI)	
`.../n<N>/aggregate/`	Cross-budget plots: regret curves, convergence probability, comparison heatmap.
`.../n<N>/history/`	Trajectory plots: best/current value, instantaneous and cumulative regret, Time to First Optimum (TTFO) distribution.
`.../n<N>/distributions/`	Budget-specific plots: regret box-plots and performance profiles.
`.../n<N>/profiles/`	Runtime profile plots: inverse profile surface/curves, $\mathbb{E}[\mathrm{CR}(T)]$ verification. Requires `suite.profile: true`.

Table 4.2: Output file organization for a representative experiment. "Normalization" replaces spaces, parentheses, and special characters with underscores or removes them.

Plot	**Output filename pattern - `PDF` format**	**Profile required**
Regret curves	`regret_curves`	No
Convergence probability	`convergence_probability`	No
Comparison heatmap	`comparison_heatmap`	No
Regret box-plots	`regret_boxplot_b{budget}`	No
Performance profile	`performance_profile_b{budget}`	No
History (best-so-far)	`history_best`	No
Instantaneous regret	`history_regret_instantaneous`	No
Cumulative regret	`history_regret_cumulative`	No
TTFO distribution	`history_ttfo_markers`	No
Runtime profile surface	`inverse_runtime_profile_surface_{alg}`	Yes
Runtime profile curves	`inverse_runtime_profile_curves`	Yes
CR profile verification	`cr_profile_verification`	Yes

Table 4.3: Plot types generated across experiments. "Profile required" denotes plots that require `suite.profile: true`

4.2 Testing strategy

The test suite follows a Test-Driven Development (TDD) discipline using a red–green–refactor workflow: tests were written prior to or alongside implementation and serve as executable specifications [Agh+23; CF23]. Coverage is measured using the `coverage` [Bc09] package with branch tracking enabled, and a minimum threshold of **50% branch coverage** is enforced in Continuous Integration (CI) via GitHub Actions. Coverage is also computed across Python 3.12, 3.13, and 3.14 in Continuous Integration (CI) for redundancy and support (the project must be compatible for `python` $\geq$ 3.12).

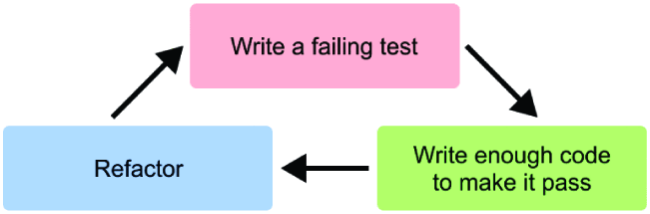


Figure 4.3: Red–Green–Refactor TDD technique. Start by writing a failing test (the red step). Next, write code that is just enough to make the written test pass (the green step). Finally, as the regularities and properties of the problem space are understood, refactors can be implemented that keeps the code clean (the blue step). The refactor step is safe since it will always be backed up by tests. [CF23]

Test organization follows a selective weighting strategy: core algorithmic components (`test_algorithms.py`, `test_metrics.py`, `test_problems.py`) and integration paths (`test_integration.py`) receive dense coverage, while infrastructure components such as CLI wiring and plot dispatch are covered at a coarser granularity.

4.3 Project Dependencies

The project is built with `hatchling` [Hat22] and managed with `uv` [Ast24], a Rust-based package and environment manager that replaces the Python standard `pip/venv`. Runtime installation: `uv sync`; development installation: `uv sync --group dev`.

Package	Min. version	Purpose
Scientific computing		
`numpy` [Har+20]	2.4.3	Array operations, random number generation
`scipy` [Vir+20]	1.16.3	Statistics, drift computations, profile integrals
`pandas` [tea20]	2.3.3	Results aggregation, tabulation, CSV export
Visualisation		
`matplotlib` [Hun07]	3.10.8	All static plot generation
`ipympl`	0.10.0	Interactive Matplotlib in notebooks
`networkx` [HSS08]	3.6.1	Graph problem instances (e.g. Petersen colouring)
Configuration and CLI		
`pyyaml` [Sc06]	6.0.3	Experiment configuration parsing
`jsonschema` [Pc12]	4.26.0	YAML schema validation
`typer` [Ram19]	0.24.1	`run_experiment` CLI
`tqdm` [Cos+26]	4.67.1	Progress bars for parallel runs
Notebook and interactive demonstration		
`jupyter` [GP21]	1.1.1	Notebook server
`ipykernel`	7.2.0	Jupyter kernel support
`ipywidgets`	8.1.8	Notebook UI widgets
Development — linting (optional)		

`ruff` [Ast22]	—	Fast Rust-based linter and formatter (E, W, F, I, B, UP rules)
Development — testing (optional)		
`pytest` [Kre+04]	—	Unit and integration test runner
`coverage` [Bc09]	—	Branch coverage reporting (≥50% enforced)
`tabulate`	—	Tabular output in test fixtures
Development — type checking (optional)		
`ty` [Ast25]	—	Rust-based static type checker (Astral)
Development — documentation (optional)		
`zensical`	—	Documentation site generator
`mkdocstrings-python`	—	Auto-generated API reference from docstrings

Table 4.4: Package dependencies for the `regret` framework (Python ≥ 3.12).

Chapter 5

Methodology

5.1 Experiment Structure

All experiments share a common infrastructure defined in the YAML configuration with schema described in Appendix E. Each configuration specifies four top-level blocks: `suite` (global settings), `problems` (benchmark functions), `algorithms` (optimizers), and `plotting` (output).

5.1.1 Repetitions and budget

Each algorithm–problem pair is run for R independent repetitions (typically 25–100) at budgets drawn from a per-suite budget list. The display horizon `budget_for_plots` is chosen to be roughly twice the expected time-to-first-optimum of the hardest algorithm on each problem, so trajectory plots show both approach and post-convergence behavior.

5.1.2 Execution mode

All suites reported in this chapter were run in `full` mode with parallelism enabled. Full mode records complete per-evaluation trajectories, which are required for the instantaneous-regret, cumulative-regret, and TTFO analyses, and for all runtime-profile computations.

5.1.3 Visualization and Table Strategy

Each suite produces plots and summary tables from the stored JSON results. Table 5.1 lists every plot type with its key, diagnostic purpose, and the suites or conditions that enable it.

Group	**Output key**	**Diagnostic purpose**	**y-axis**
Aggregate	`regret_curves`	Mean simple regret vs. budget; crossover budget between algorithms	log
	`convergence_probability`	P(opt) vs. budget; separates partial from full convergence	linear
	`comparison_heatmap`	$\log_{10}$(mean regret) colour-coded over all alg–budget cells	—
Distribution	`regret_boxplots`	Run-level spread of simple regret; skewness from non-converging runs	linear

	`performance_profile`	Empirical CDF of regret on log axis; steep rise = reliable convergence	log
Trajectory	`history_best`	Monotone incumbent fitness with f^* reference	linear
	`regret_instantaneous`	IR of evaluated solution; fluctuates when worsening moves accepted	log
	`regret_instantaneous_best`	IR of incumbent (monotone non-increasing); non-zero floor = saturation	log
	`regret_cumulative`	Integral of current-solution IR; penalises slow convergence	linear
	`regret_cumulative_best`	Integral of incumbent IR; key signal in barrier-crossing suites	linear
	`ttfo_distribution`	Jittered TTFO samples per algorithm; conditioned on convergence	linear
Runtime profile (suites 03, 08, 12)	`inverse_runtime_profile_surface`	$P(\tau_v \leq T)$ heat map over (T, v); saturation frontier visible	—
	`inverse_runtime_profile_curves`	$P(\tau_v \leq T)$ vs. T at $f^* \times \{0.25, 0.5, 0.75, 0.95\}$	linear
	`cr_profile_verification`	Overlay of direct and profile-derived $\mathbb{E}[\mathrm{CR}(T)]$	linear
Summary	per-problem table	Mean, median, SD, IQR, 95% CI, $P(\mathrm{opt})$, run count per algorithm	—
	aggregate table	Across-problem mean regret, mean $P(\mathrm{opt})$, best-count per algorithm	—

Table 5.1: Plot types produced by the experimental pipeline. The "Output key" column matches the YAML `plots` sub-block name. All trajectory and distribution plots are generated at `budget_for_plots`; aggregate plots pool across all budgets in the suite list. IR = instantaneous regret; CR = cumulative regret.

For hill-climbing algorithms (RLS, $(1+1)$-EA) the current-solution and incumbent IR variants coincide because no worsening moves are accepted; for SA and RLSExploration they diverge during phases of active exploration, making the current-solution variant a direct diagnostic of temperature-driven behaviour. Cumulative regret is always plotted on a linear y-axis: its slope during the trapped phase grows linearly with time, and a log axis would suppress that signal.

5.1.4 Suite Numbering and Roles

Suites are numbered 01–13. Table 5.2 gives a complete reference; the methodological context for runtime-profile suites (03, 08, 12) is developed in Chapter 3.

Suite	Role	Problems	Algorithms	R	Max budget	Profile
01	Baseline comparison	6 pseudo-boolean	RLS, RLSExpl, EA, SA-Log-d2.0	50	10 000	No
02	Extended benchmark	9 pseudo-boolean + combinatorial	same 4 families (with overrides)	50	20 000	No
08	Tail-sum verification	OneMax, LeadingOnes	RLS, $(1+1)$-EA	100	10 000	**Yes**
03	NK ruggedness (6×5 grid)	5 NK ($k\in\{1,2,4,6,8\}$, $n=20$)	RLS, EA, 6×SA-Log	25	50 000	**Yes**
04	Trap: deception width × rate	4 Trap ($k\in\{5,8,10,20\}$)	4×EA + SA-Log-d4.0	25	10 000	No
05	Jump: mutation rate / plateau onset	5 Jump ($k\in\{2,3,5,7,10\}$)	3×EA (fixed + adaptive k/n)	25	20 000	No
06	Plateau: neutral walk × rate	4 Plateau ($k\in\{2,5,10,20\}$)	3×EA + SA-Log-d4.0	30	20 000	No
07	HIFF scaling	3 HIFF ($n\in\{32,64,128\}$)	RLS, 4×EA, SA-Log-d4.0	25	50 000	No
09	Jump: plateau duration (extended)	5 Jump (same as 05)	2×EA (fixed + adaptive)	25	50 000	No
10	Barrier archetype comparison	Jump/Trap/Plateau ($k=5$, $n=100$)	4×EA ($0.5/n$–$5/n$)	30	50 000	No
11	SA cooling diagnostics	OneMax, LeadingOnes	RLS, 6×SA-Log, SA-Lin, SA-Exp	50	10 000	No
12	NK saturation	5 NK ($k\in\{0,2,4,6,8\}$, $n=20$)	RLS, $(1+1)$-EA, SA-Log-d4.0	25	50 000	**Yes**
13	Regret scaling with n	4 OneMax ($n\in\{25,50,100,200\}$)	$(1+1)$-EA at rate $1/n$	60	10 000	No

Table 5.2: Overview of all 13 experimental suites. R = runs per algorithm–problem–budget combination. "Profile" indicates whether `suite.profile: true` is set. Budget values are in evaluations; "max" is the largest value in the suite budget list.

5.2 Baseline Suites

5.2.1 Suite 01: Algorithm-Family Comparison

Suite 01 establishes a reference comparison of four algorithm families across six pseudo-boolean problems of increasing structural difficulty: Randomized Local Search (RLS), RLSExploration, the $(1+1)$-EA, and SA with logarithmic cooling at $d = 2.0$. SA is reduced to a single representative schedule here; the effect of varying the schedule parameter is deferred to suites 03 and 11.

The six problems are: OneMax ($n = 100$), LeadingOnes ($n = 100$), TwoMax ($n = 100$), Jump-3 ($n = 100$, $k = 3$), Trap-10 ($n = 100$, $k = 10$), and BinVal ($n = 50$). BinVal uses $n = 50$ rather than $n = 100$ because its exponential fitness structure already produces large fitness gaps and rapid convergence at this size; the `budget_for_plots` of 500 reflects this. The 50 runs per combination are evaluated at six budgets: $\{200, 500, 1000, 2000, 5000, 10000\}$. Pairwise statistical annotations use RLS as the reference algorithm on box-plots, performance profiles, and TTFO distributions.

5.2.2 Suite 02: Extended Pseudo-Boolean Benchmark

Suite 02 extends the baseline to nine problems, adding Jump-5, Trap-5, Plateau-10, HIFF-64, and a combinatorial instance (Petersen-3-Color). The algorithm set is identical to suite 01. Budgets range from 500 to 20 000 over 50 runs. Profile analysis is disabled.

Three problems receive problem-specific algorithm overrides (full parameter values are listed in Table 5.3 in the appendix). For RLSExploration on Jump-3, Jump-5, and Trap-5, the exploration rate is raised to $\varepsilon = 0.05$ with decay disabled, maintaining persistent non-improving moves on barrier problems. For SA, the cooling parameter is tripled to $d = 6.0$ on Jump-5 and Trap-5, raising acceptance probability for worsening moves throughout the run. Both modifications target the same objective: improving escape from local or deceptive optima at moderate budget.

Algorithm	**Problem**	**Parameter**	**Default**	**Override**
RLSExploration	Jump-3	ε, decay	0.01, on	0.05, off
	Jump-5	ε, decay	0.01, on	0.05, off
	Trap-5	ε, decay	0.01, on	0.05, off
SA-Log-d_2.0	Jump-5	d	2.0	6.0
	Trap-5	d	2.0	6.0

Table 5.3: Problem-specific algorithm overrides in suite 02. For RLSExploration the default is $\varepsilon = 1/n = 0.01$ with exponential decay ($\varepsilon_t = \varepsilon/t$); for SA the default cooling parameter is $d = 2.0$.

5.3 Verification Suite

5.3.1 Suite 08: Runtime Profile Verification

Suite 08 validates the tail-sum identity (Chapter 3)

$$\mathbb{E}[\mathrm{CR}(T)] = \sum_{v=1}^{f^*} \sum_{t'=1}^{T} \left[1 - P(\tau_v \le t')\right]$$

on OneMax and LeadingOnes ($n = 100$) using RLS and the $(1+1)$-EA. With $R = 100$ independent runs (the highest run count in the study) Monte-Carlo variance in $\widehat{\mathbb{E}}[\mathrm{CR}(T)]$ is suppressed to the point where any visible gap between trajectory-derived and profile-derived curves in the verification plot can be attributed to numerical discretization rather than sampling noise. The `budget_for_plots` for LeadingOnes is set to the maximum

available (10 000) to capture the full cumulative regret ramp, which extends further than on OneMax due to its $\Theta(n^2)$ expected convergence time. Profile analysis is enabled; pairwise annotations use RLS as the reference.

5.4 Exploratory Analysis of Regret and Runtime

The exploratory suites address four themes: barrier structure and mutation rate (suites 04–06, 09–10), NK ruggedness and SA cooling (suites 03, 11–12), hierarchical structure, and scaling (suites 07, 13).

5.4.1 Suite 03: NK Landscape Ruggedness

Suite 03 crosses six logarithmic SA schedules ($d \in \{0.5, 1.0, 2.0, 4.0, 8.0, 16.0\}$) with five NK instances ($k \in \{1, 2, 4, 6, 8\}$, $n = 20$, seed 42). RLS and the $(1+1)$-EA are included as zero-temperature baselines so that the effect of temperature-driven acceptance can be separated from plain hill-climbing; on low-epistasis instances SA may in fact underperform hill-climbing, which the baselines make visible.

All SA entries carry empty `by_problem` blocks, forming a clean 6×5 grid over cooling parameter and epistasis level without per-problem tuning. The restriction to $n = 20$ is imposed by the exact-optimum computation used in the NK implementation. Profile analysis is enabled, and the budget extends to 50 000 because high-epistasis instances may approach saturation only late in the run.

5.4.2 Suite 04: Trap Function Mutation Rate

Suite 04 evaluates Trap with four $(1+1)$-EA mutation rates ($0.5/n$, $1/n$, $2/n$, $5/n$) and SA-Log-d_4.0 across four deception widths ($k \in \{5, 8, 10, 20\}$, $n = 100$). The EA rates are fixed across all problems; SA uses $d = 4.0$ by default with a single override to $d = 8.0$ on Trap-k20. Trap-k1 is excluded because all algorithms are uniformly driven to the deceptive attractor, producing uninformative flat results. The chosen EA rates are shared with suites 05 and 06 to support cross-suite comparison of regret shape across barrier types. Profile analysis is disabled.

5.4.3 Suite 05: Jump Function Mutation Rate

Suite 05 studies five Jump instances ($k \in \{2, 3, 5, 7, 10\}$, $n = 100$) under three $(1+1)$-EA variants: fixed rates $1/n$ and $2/n$, and a problem-adaptive rate k/n set via per-problem overrides. The lower rate $0.5/n$ is omitted because it offers no advantage over $1/n$ for crossing a k-bit gap. Budgets range from 500 to 20 000. Instantaneous regret is plotted on a log scale, while cumulative regret uses a linear scale so that the trapped-phase slope remains visible. Profile analysis is disabled; the longer-horizon Jump study is deferred to suite 09.

5.4.4 Suite 06: Plateau Function Mutation Rate

Suite 06 mirrors suite 05 on Plateau instances ($k \in \{2, 5, 10, 20\}$, $n = 100$) to compare cumulative regret trajectory shape across barrier types at structurally equivalent widths. The algorithm set is three $(1+1)$-EA variants at rates $1/n$, $2/n$, and $5/n$, plus SA-Log-d_4.0.

The adaptive k/n rate from suite 05 is omitted: it was motivated by the need to maximize the probability of exactly k simultaneous flips for the Jump gap, a rationale that does not apply to the neutral walk on Plateau. SA is added to assess whether temperature-driven acceptance confers any advantage on a landscape where the neutral zone carries no fitness penalty. The shared values $k = 2$ and $k = 5$ support direct comparison with suite 05. Profile analysis is disabled.

5.4.5 Suite 07: Hierarchical IFF Scaling

Suite 07 evaluates HIFF at $n \in \{32, 64, 128\}$ under RLS, four $(1+1)$-EA variants at rates $\{0.5/n, 1/n, 2/n, 3/n\}$ (exact per-problem decimal values listed in Table 5.4), and SA-Log-d_4.0 at a fixed $d = 4.0$. RLS is expected to stagnate at the first hierarchical level for $n \geq 64$ since a single bit-flip disrupts any uniform sub-block, anchoring

the comparison. For $n = 128$, `budget_for_plots` is set to the maximum available (50 000); convergence is not expected, and the unsolved trajectory provides an informative contrast to the partially solved $n = 32$ and $n = 64$ cases. Profile analysis is disabled.

Algorithm	$n = 32$	$n = 64$	$n = 128$
EA-mu_0.5-by-n	$1/64 \approx 0.01563$	$1/128 \approx 0.00781$	$1/256 \approx 0.00391$
EA-mu_1-by-n	$1/32 \approx 0.03125$	$1/64 \approx 0.01563$	$1/128 \approx 0.00781$
EA-mu_2-by-n	$2/32 \approx 0.06250$	$2/64 \approx 0.03125$	$2/128 \approx 0.01563$
EA-mu_3-by-n	$3/32 \approx 0.09375$	$3/64 \approx 0.04688$	$3/128 \approx 0.02344$

Table 5.4: Per-problem mutation rates in suite 07. Values are set via `by_problem` overrides to exact decimal representations of $\{0.5/n, 1/n, 2/n, 3/n\}$ at each problem size.

5.4.6 Suite 09: Jump Plateau Duration

Suite 09 uses the same five Jump instances and two EA variants as suite 05 (rates $1/n$ and k/n with identical per-problem overrides), extending the budget list to $\{1000, 5000, 10000, 25000, 50000\}$ and raising `budget_for_plots` to 50 000 for the larger k values. Where suite 05 captures plateau onset for small k, suite 09 captures plateau duration across all k values and, for small k at the adaptive rate, potentially the plateau exit. Profile analysis is disabled.

5.4.7 Suite 10: Barrier Structure Comparison

Suite 10 compares Jump-k5, Trap-k5, and Plateau-k5 at $n = 100$ under four $(1+1)$-EA mutation rates: $0.5/n$, $1/n$, $2/n$, and $5/n$, applied uniformly across all three barrier types with budgets to 50 000. Since all three barriers share $\mathbb{E}[\text{TTFO}] = \Theta(n^k)$ at rate $1/n$, TTFO distributions cannot separate them when $P(\text{opt}) \approx 0$; cumulative regret trajectories accumulate differently depending on the fitness gradient before and during the barrier, and testing whether this distinction is visible at $n = 100$, $k = 5$ is the suite's primary purpose. Pairwise annotations use `(1+1)-EA-mu_1-by-n` as the reference. Profile analysis is disabled.

5.4.8 Suite 11: SA Cooling Schedule Diagnostics

Suite 11 evaluates six logarithmic SA schedules ($d \in \{0.5, 1.0, 2.0, 4.0, 8.0, 16.0\}$), one linear schedule (SA-Lin), one exponential schedule (SA-Exp), and RLS as a zero-temperature reference on OneMax and LeadingOnes ($n = 100$). The linear and exponential variants use implementation defaults (linear: $T_0 = 1.0$, $T_f = 0.01$, max_iter $= 10\,000$; exponential: $T_0 = 1.0$, $\alpha = 0.95$); both reach their minimum temperature within the budget (exponential effectively by $t \approx 90$), after which SA reduces to a near-deterministic hill-climber. This structural distinction between collapsing and sustaining schedules is the primary hypothesis the suite is designed to expose via instantaneous regret trajectories. Pairwise annotations use RLS as the reference so that any SA configuration underperforming plain hill-climbing is directly identifiable. Profile analysis is disabled.

5.4.9 Suite 12: NK Saturation

Suite 12 examines five NK instances ($k \in \{0, 2, 4, 6, 8\}$, $n = 20$, seed 42) under RLS, the $(1+1)$-EA, and SA-Log-d_4.0 with per-k cooling parameter escalation (values in Table 5.5). The landscape realizations at $k \in \{2, 4, 6, 8\}$ are identical to suite 03, enabling direct comparison. The addition of $k = 0$ (fully separable) provides a saturation baseline. Unlike suite 03, SA is calibrated per problem rather than forming a grid; suite 12 therefore asks whether adequate temperature maintenance changes the saturation outcome rather than how the cooling–epistasis interaction behaves across a full parameter sweep. Profile analysis is enabled; the `cr_profile_verification` plot is omitted because the NK fitness function is continuous and the integer-grid profile identity does not apply exactly.

NK instance	NK-k0	NK-k2	NK-k4	NK-k6	NK-k8
d	4.0	4.0	8.0	8.0	12.0

Table 5.5: SA-Log cooling parameter d per NK instance in suite 12. Values increase monotonically with epistasis k to compensate for deeper local optima.

5.4.10 Suite 13: Regret Scaling with Problem Size

Suite 13 runs the $(1+1)$-EA at rate $1/n$ on four OneMax instances ($n \in \{25, 50, 100, 200\}$) over 60 runs (the second-highest run count in the study) to produce empirical $\mathbb{E}[\mathrm{CR}(T)]$ estimates suitable for fitting a scaling exponent. Using a single algorithm isolates the scaling behavior of the regret metric from algorithm-to-algorithm differences. `budget_for_plots` values are set somewhat above $5n \log_2 n$ so that the full post-convergence plateau is visible in trajectory plots. Profile analysis is disabled.

5.5 Cross-Suite Design Relationships

Several structural relationships between suites affect which results can be meaningfully compared. Table 5.6 summarizes the three principal links.

Relationship	**Suites**	**Shared**	**Differs**
Barrier types at equal rates	04, 05, 06, 10	EA rates $1/n$ and $2/n$; problems at same nominal k	Barrier type (Trap/Jump/Plateau); additional rates per suite
Jump at two time horizons	05, 09	All 5 Jump instances; EA variants at $1/n$ and k/n	Budget list and `budget_for_plots` (onset vs. duration)
NK at two SA designs	03, 12	$n = 20$, seed 42; shared $k \in \{2, 4, 6, 8\}$; identical landscapes	Suite 03: 6×5 d-grid, no tuning; Suite 12: per-k d escalation, adds $k = 0$

Table 5.6: Principal cross-suite design relationships.

Suites 04, 05, and 06 share the fixed rates $1/n$ and $2/n$, making Trap-k, Jump-k, and Plateau-k results directly comparable at equal k; suite 10 consolidates this at $k = 5$ across all four EA rates simultaneously. Suites 05 and 09 are identical in problem and algorithm design, differing only in budget horizon: suite 05 captures plateau onset, suite 09 captures plateau duration. Suites 03 and 12 share landscape realizations at $k \in \{2, 4, 6, 8\}$ but differ in SA design: suite 03 provides the clean grid for studying the cooling–epistasis interaction, while suite 12 provides the per-k calibrated view of saturation.

Chapter 6

Evaluation and Discussion of Results

6.1 Overview

6.2 Baseline Suites

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6.2.1.2 Regret Distributions and TTFO

6.2.1.3 Trajectory Analysis

6.2.2 Suite 02: Extended Pseudo-Boolean Benchmark

6.2.2.1 Aggregate Regret and Convergence

6.2.2.2 Effect of Algorithm Overrides

6.2.2.3 Trajectory Analysis

6.3 Verification Suite

6.3.1 Suite 08: Runtime Profile Verification

6.3.1.1 Tail-Sum Identity: Direct vs. Profile-Derived $\mathbb{E}[\mathrm{CR}(T)]$

6.3.1.2 Inverse Runtime Profile Surfaces and Curves

6.4 Exploratory Analysis

6.4.1 Barrier Structure and Mutation Rate

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6.4.1.2 Suite 05: Jump – Mutation Rate and Plateau Onset

6.4.1.3 Suite 06: Plateau – Neutral Walk and SA

6.4.1.4 Suite 09: Jump – Plateau Duration

6.4.1.5 Suite 10: Barrier Archetype Comparison at k=5

6.4.2 NK Ruggedness and SA Cooling

6.4.2.1 Suite 03: Cooling Schedule × Epistasis Grid

6.4.2.2 Suite 11: SA Cooling Schedule Diagnostics

6.4.2.3 Suite 12: NK Saturation

Chapter 7

Conclusion and Future Work

7.1 Conclusion

7.2 Future Work

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Appendix A

Evolutionary Algorithms, Extended

A.1 Exact Runtime Formulae

For mutation probability p, the exact expected optimization time on LeadingOnes is [BDN10]

$$\mathbb{E}[T_p] = \frac{1}{2p^2}\big((1-p)^{-n+1} - (1-p)\big),$$

minimized at $p \approx 1.59/n$, giving $\approx 0.77n^2$. For OneMax, the sharp expansion at $p = 1/n$ is [Hwa+18]

$$\mathbb{E}[T] = e\,n\ln n - 1.8925\ldots n + \tfrac{e}{2}\ln n + O(1).$$

More generally, for $p = c/n$ the expected time on any linear function is $(1 \pm o(1))\frac{e^c}{c}\,n\ln n$, minimized at $c = 1$ [Wit13].

A.2 Implementation-Aware Variants

A practical observation of Pinto and Doerr [PD18] is that roughly a $1/e$ fraction of all $(1+1)$ EA iterations flip *no* bit (the offspring equals the parent), yet standard analyses charge a full function evaluation for each such iteration. Real implementations avoid these wasted evaluations by one of two strategies.

Resampling variant – $(1+1)$ EA$_{>0}$. The mutation strength ℓ is drawn from the *conditional* binomial distribution $\mathrm{Bin}_{>0}(n,p)$, i.e., $\mathrm{Bin}(n,p)$ conditioned on $\ell > 0$. This effectively skips any iteration that would flip nothing.

Shift variant – $(1+1)$ EA$_{0\to1}$. If $\ell = 0$ is sampled, it is replaced by $\ell = 1$, shifting all probability mass from zero flips to exactly one flip.

Both variants are analyzed in full in [PD18; JZ11]. The key runtime consequences with $p = 1/n$ are summarized in Table A.1.

Algorithm	OneMax	LeadingOnes
$(1+1)$ EA	$e\,n\ln n + O(n) \approx 2.718\,n\ln n$	$\approx 0.859\,n^2$
$(1+1)$ EA$_{>0}$	$\approx 1.718\,n\ln n + O(n)$	$\approx 0.543\,n^2$
$(1+1)$ EA$_{0\to1}$	$\approx 1.359\,n\ln n + O(n)$	$\approx 0.517\,n^2$
RLS	$nH_n \approx n\ln n$	$\approx 0.5\,n^2$

Table A.1: Expected optimization times at $p = 1/n$ for the $(1+1)$ EA and its implementation-aware variants. RLS is included for reference. Bounds for the variants follow from [PD18; BDN10; Wit13].

Resampling or shifting yields a 35–47% reduction in function evaluations on OneMax and roughly 37–40% on

LeadingOnes at $n = 600$, as confirmed empirically in [PD18]. For the (1+1) $\text{EA}_{>0}$, the expected runtime on any linear function with $p = c/n$ becomes $(1 \pm o(1)) \frac{e^{c-1}}{c} n \ln n$ [PD18], a multiplicative $(1 - e^{-c})$ improvement over the standard algorithm.

A.3 General Form - '$(\mu + \lambda)$-EA'

A.4 Genetic Algorithms

A.4.1 Crossover

A.5 Parameters and Operators

A.5.1 Adaptive Mutation and Crossover Rates

A.5.2 Selection pressure

Appendix B

Supplementary Optimization Background

B.1 Oracle Tiers: First- and Second-Order Methods

The three tiers of oracle access form an information hierarchy that governs both the choice and the efficiency of applicable algorithms.

First-Order Methods. First-order methods additionally access the gradient $\nabla f(x) \in \mathbb{R}^d$, which encodes the direction of steepest ascent at x. For a differentiable function $f : \mathbb{R}^d \to \mathbb{R}$, gradient-based methods exploit this to define descent directions and form the backbone of classical continuous optimization [NW06]. When f is convex but non-differentiable, the gradient generalizes to the *subdifferential* $\partial f(x)$, the set of all *subgradients* g satisfying

$$f(y) \geq f(x) + g^\top (y - x) \quad \forall y \in \mathbb{R}^d.$$

Subgradient methods replace the gradient with any element of $\partial f(x)$, recovering convergence under convexity alone, albeit at rates slower than those attainable with smooth gradients [Ber09].

Second-Order Methods. Second-order methods further access the Hessian $\nabla^2 f(x) \in \mathbb{R}^{d \times d}$, which captures local curvature. Methods that utilize or approximate the Hessian, such as Newton and quasi-Newton schemes, can achieve super-linear or quadratic convergence near a solution [NW06], at the cost of maintaining a $d \times d$ matrix. Together, the three tiers define distinct levels of oracle access; restricting access progressively from second-order to zeroth-order culminates in the black-box model where only function values are observable.

B.2 Landscape Properties: Convexity and Smoothness

Convexity. A function $f : \mathbb{R}^d \to \mathbb{R}$ is *convex* if, for all $x, y \in \mathbb{R}^d$ and $\lambda \in [0,1]$,

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$$

[Ber09, Def. 1.1.1]. For a convex function over a convex domain, every local minimum is also a global minimum [Ber09, Prop. 1.3.2]. A differentiable function is *strongly convex* with parameter $\mu > 0$ if

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \tfrac{\mu}{2} \|y - x\|^2,$$

which is one of the standard equivalent characterizations of strong convexity [Nes+18, Thm. 2.1.9]. Strong convexity implies a unique global minimizer [Nes+18, Thm. 2.1.8]. Moreover, if f is also L-smooth, then gradient descent with a suitable fixed step size converges linearly [Nes+18, Thm. 2.1.15].

Smoothness and Lipschitz Continuity. A differentiable function f is *L-smooth* if its gradient is Lipschitz continuous with constant $L > 0$:

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \quad \forall x, y \in \mathbb{R}^d.$$

Smoothness bounds the curvature of f from above and is the standard assumption under which gradient descent achieves an $O(1/T)$ convergence rate for convex objectives [NW06]. The objective f itself is *Lipschitz continuous* with constant G if $|f(x) - f(y)| \leq G\|x - y\|$, a weaker condition that enables subgradient convergence without differentiability. Neither property is assumed in the black-box pseudo-Boolean setting studied in this project.

B.3 Hajék's Convergence Condition for Simulated Annealing

Modelling simulated annealing as a time-inhomogeneous Markov chain on the state space $\mathcal{X}$, Hajék [Haj88] established the following sharp necessary and sufficient condition for convergence in probability to the global optimum.

Theorem 5 (Hajék 1988)**.** *Let d^* denote the* depth *of the deepest local minimum that is not globally optimal, defined as the minimum energy that must be crossed to escape it. A cooling schedule $T(t) > 0$ guarantees that the distribution of the current state converges to a distribution supported on the global optima if and only if*

$$\sum_{t=1}^{\infty} \exp\left(\frac{-d^*}{T(t)}\right) = +\infty.$$

For the logarithmic schedule $T(t) = d/\log(t+1)$, the condition reduces to the requirement $d \geq d^*$: the scaling constant must be at least as large as the hardest energy barrier in the landscape [Haj88; BT93]. Logarithmic cooling is therefore theoretically optimal in the sense that it is the fastest schedule satisfying the convergence condition.

In practice, d^* is unknown for general combinatorial problems, and logarithmic cooling is often too slow to be useful within a finite evaluation budget. Exponential cooling $T(t) = T_0 \cdot \alpha^t$ ($\alpha \in (0,1)$) and linear cooling $T(t) = \max(T_f, T_0 - (T_0 - T_f)\, t/t_{\max})$ cool faster but violate the condition once the temperature reaches zero, at which point SA reduces to a deterministic hill-climber. The choice of schedule therefore encodes an explicit trade-off between the convergence guarantee and the practical computational budget.

B.4 Plateau Function: Runtime Analysis

Antipov and Doerr [AD21] establish that for any unbiased mutation operator the expected runtime of the $(1+1)$ EA on $\textsc{Plateau}_k$ is

$$\mathbb{E}[T] = \frac{n^k}{k!\,\Pr[1 \leq \alpha \leq k]}\,(1 + o(1)),$$

where α is the number of bits flipped per step. This expression depends only on the probability of flipping between one and k bits, not on any other property of the operator. The optimal mutation rate under standard bit mutation is approximately $k/(en)$, a factor of $\Theta(k)$ above the canonical $1/n$ recommendation, mirroring the intuition that wider plateaus require more aggressive perturbation to exit. This result is directly relevant to the suite 06 and suite 10 experimental analyses, where the contrast between Jump and Plateau regret trajectories is examined at shared k values.

Appendix C

Extended Black-Box Complexity Theory

C.1 Las Vegas vs. Monte Carlo Black-Box Complexity

In the standard (unrestricted, unbiased) black-box models, the *Las Vegas* complexity (the expected number of queries until an optimal solution is first found) and the *Monte Carlo* complexity (the minimum budget T such that the optimum is found with probability at least $1-p$ for a given $p>0$) are essentially equivalent up to constant factors via Markov's inequality: any Las Vegas algorithm with expected runtime T finds the optimum within T/p queries with probability at least $1-p$.

In elitist models, however, this equivalence breaks down severely. Doerr and Lengler [DL15] show that the $(1+1)$ elitist Las Vegas and Monte Carlo complexities of OneMax can be exponentially far apart. The reason is structural: an elitist algorithm cannot perform restarts, because a fresh random starting point will almost surely have lower fitness than the current incumbent and will therefore be rejected. Without restarts, a run that fails to find the optimum is trapped, and there is no mechanism to recover. Consequently, failure probabilities do not decay geometrically with additional budget in the same way as for non-elitist algorithms.

Formally, the *p-Monte Carlo black-box complexity* of a problem class $\mathcal{F}$ with respect to an algorithm class $\mathcal{A}$ is

$$\min_{A\in\mathcal{A}} \min\{T\in\mathbb{N} : \forall f\in\mathcal{F},\, \mathbb{P}(\tau_{f^*}\le T)\ge 1-p\}.$$

The Las Vegas complexity is recovered in the limit by Markov's inequality when the two notions coincide, but for the $(1+1)$ elitist Las Vegas complexity of OneMax, Doerr and Lengler establish only that it is $O(n\log n)$ (matching the unary unbiased lower bound) while leaving the exact constant open; the Monte Carlo complexity is $\Theta(n)$, a separation of a full $\log n$ factor [DL15].

For the algorithms studied in this project, this distinction does not affect the experimental results directly: all runtime and regret analyses use expected quantities over independent runs, which corresponds to the Las Vegas framework.

C.2 Yao's Principle in Restricted Models

Yao's Minimax principle in 2.2.2.2 applies directly only to algorithm classes whose convex hull contains the randomized algorithm of interest. For memory-restricted and elitist models this containment fails because some randomized algorithms in these classes are not convex combinations of deterministic ones. The canonical example is RLS itself: every deterministic $(1+1)$ memory-restricted ranking-based algorithm that ever rejects a search point will enter an infinite loop on OneMax with positive probability when the input is chosen uniformly at random, giving infinite expected runtime, whereas RLS optimizes OneMax in $\Theta(n\log n)$ expected queries [DL15].

Doerr and Lengler [DL15] resolve this by applying Yao's principle to the strictly larger class $\mathcal{A}'$ of all comparison-based $(1+1)$ algorithms with access to their complete search history. Every memory-restricted ranking-based algorithm can be simulated in $\mathcal{A}'$, so a lower bound on the $\mathcal{A}'$-complexity implies the same lower bound for the original class. The information- theoretic argument then proceeds as in the unrestricted case: after $i-1$ queries the oracle has revealed at most one bit of information per query, so at most 2^{i-1} candidate optima have been eliminated, giving

$$\mathbb{P}(\text{optimum found by query } i) \leq 2^{i-n},$$

and summing yields $\mathbb{E}[\tau_{f^*}] \geq n-1$.

Appendix D

Related Works

D.0.1 Related Work: Regret in Bayesian and Surrogate-Based Optimization

D.0.2 Related Work: Regret-Inspired Analysis of Evolutionary Algorithms

D.0.3 Related Work: Bandit-Inspired Metaheuristics

Appendix E

Configuration Schema

Listing E.1: Schema for the configuration files to describe experiments

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